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S1. Construction of BrainTumor46K dataset

We assembled the BrainTumor46K dataset through systematic curation from 36 online sources and 12 collaborating medical centers. To guide the search and data collection processes, we first established a comprehensive set of brain tumor keywords, including 12 major brain tumor types and all subtypes, along with their synonyms and related terms. Next, we leveraged retrieval-augmented generation (RAG) and prompt engineering techniques to design an automated preprocessing pipeline, see Supplementary Figure 1. This pipeline transformed the curated raw datasets into structured formats of $\{Patient_ID: \text{multi-parametric 3D MRI volumes (or 2D MRI slice), pathological diagnosis label, optional radiology report (or text description)}\}$ for model training. We also expanded our datasets by incorporating data from healthy individuals. For the multi-parametric MRI scans, we considered T1-weighted (T1), T1 contrast-enhanced (T1c), T2-weighted (T2), and T2-FLAIR (FLAIR), which are commonly used in brain tumor diagnosis.

S1.1. Dataset construction from online repositories

We compiled our dataset from diverse online repositories, categorized based on their inherent structure and the required preprocessing complexity. The first category well-structured sources of medical imaging data, such as established repositories (e.g., BraTS2023,^{1,2,3,4,5,6} TCIA,^{7,8,9,10,11,12,13,14,15,16,17,18,19,20} Kaggle,^{21,22,23,24} Cttisus²⁵) as well as specialized medical platforms like Radiopaedia.²⁶ This category encompasses sources that provide readily accessible, structured medical imaging data (including 3D multi-parametric MRI volumes or 2D slices), which are typically paired with pathological labels. Depending on the source, additional information, such as patient metadata or radiology reports, may also be available. The second category involves non-dedicated imaging sources requiring advanced figure/text extraction, such as PubMed Central scientific archives.²⁷ This platform, not originally designed for medical imaging storage, necessitated specialized pipelines for isolating figures/text from heterogeneous formats (e.g., publication figures, video stills) and reconciling fragmented clinical annotations. In the following, we detail the distinct data pre-processing pipelines developed for each category.

Pre-processing pipeline for structured repositories and pre-curated datasets. Datasets in this category are highly heterogeneous, encompassing various sources with differing image formats, annotations, and metadata. These datasets range from simplified collections (where each patient is represented by a single 2D MRI slice with a slice-level pathological label) to comprehensive datasets containing multi-parametric 3D MRI sequences, tumor segmentation masks, patient metadata, and pathological labels. For datasets providing only tumor classification labels, we processed the data into a standardized structure of the form: $\{Patient_ID: \text{3D MRI volumes (or 2D MRI slice), pathological label}\}$. The resulting standardized data were subsequently used for tumor classification tasks. For datasets containing 3D multi-parametric MRI scans (or 2D MRI slice), segmentation masks, and metadata (e.g., BraTS,^{1,2,3,4,5,6} TCIA^{7,8,9,10,11,12,13,14,15,16,17,18,19,20} and Kaggle datasets^{21,22}), we first applied the Brainnetome Atlas²⁸ to localize functional brain regions based on the tumor masks. We then leveraged all available metadata (including imaging modality, gender, diagnostic label, and tumor description) to generate comprehensive medical reports via GPT-4o prompted with RAG techniques (see Supplementary Figure 1a1). For certain web-sourced datasets, such as Radiopaedia,²⁶ which include multi-parametric MRI scans and accompanying radiology reports, we removed irrelevant information and reformatted the data into a consistent structure: $\{Patient_ID: \text{3D MRI volumes, pathological label, radiology report}\}$.

This comprehensive pre-processing pipeline enabled the assembly of a diverse cohort comprising (see Supplementary Table 2): 1) 6,957 brain tumor patients with 3D multi-parametric MRI scans, radiology reports, and associated pathological labels; 2) 876 brain tumor patients with 3D multi-parametric MRI scans and pathological labels only; 3) 233 patients with 2D MRI slice-text description; and 4) 3,178 healthy control patients with multi-parametric 3D MRI scans. Apart from the datasets detailed above, this preprocessing pipeline also incorporated subjects from additional repositories, such as Br35H,²¹ Brain Tumor Classification Dataset,²⁴ and Brain Tumor MRI Images 44 Classes.²³ However, as these repositories only contain 2D MRI slices with associated pathological labels and lack patient identifiers, they were not included in our dataset summary here (see Supplementary Table 2).

Pre-processing pipeline for unstructured web-based resources. Web-crawled brain tumor data from platforms such as PubMed Central,²⁷ ImageCLEF²⁹ often contain noisy 2D MRI scans and misaligned or inconsistent text descriptions. To address these challenges, we developed a comprehensive data preprocessing pipeline that transforms raw, heterogeneous web data into a structured format: $\{Patient_ID: \text{2D MRI slice, text description, pathological label}\}$, see Supplementary Figure 1a2.

For scientific articles from PubMed Central, we systematically retrieved 2D brain MRI slices by querying pre-defined brain tumor keywords, associating each image with both “golden captions” and inline textual mentions as candidate descriptions. However, these figures frequently contain composite images with multiple subfigures and inconsistent captioning. To resolve this, we developed a three-stage refinement protocol: Firstly, we applied a Faster R-CNN object detector,³⁰ trained on the MedICaT dataset,³¹ to perform subfigure separation and irrelevant images filtering. Secondly, we employed GPT-4o with specialized prompts for accurate subcaption separation. Finally, we established a precise one-to-one subfigure-caption correspondence through bounding box coordinate alignment.³² ImageCLEF data required minimal processing, with only irrelevant images filtered, as slice-text pairs were directly utilized.

In total, our curated dataset includes 24,333 unique subjects: 23,849 from PubMed Central (116,296 2D MRI slice-text pairs), and 484 from ImageCLEF (4,631 2D MRI slice-text pairs).

S1.2. Dataset construction from medical institutions

To address the scarcity of rare brain tumor types/subtypes, **the need for a prospective validation study** and the limited availability of expert-curated radiology reports in online repositories, we curated data from twelve collaborating hospitals (Supplementary Table 1). For each patient with a pathologically confirmed diagnosis, we collected de-identified multi-parametric MRI scans (T1, T1c, T2, FLAIR) that are commonly used for CNS tumor diagnosis, along with de-identified radiology reports and associated pathological diagnoses. In total, we gathered multi-modal data from 8,733 patients at the primary Xiangya Hospital, 1,329 patients from 11 other independent institutions, **and 723 patients in the prospective validation cohort**. Demographic and statistical summaries of these medical centers are provided in Figure 1 and Supplementary Table 1, 3.

The original institutional radiology reports were in Chinese and often included descriptions of other modalities (e.g., Magnetic Resonance Angiography (MRA), Magnetic Resonance Spectroscopy (MRS), Blood Oxygen Level Dependent Imaging (BOLD) signals; Supplementary Table 5). Thus, we implemented a four-stage standardization protocol to process these reports (Supplementary Figure 1c). Firstly, we used regular-expression filters to eliminate non-T1/T1c/T2/FLAIR content. Secondly, we developed a clinician-validated bilingual lexicon (Chinese-English) to accurately map medical terms, translating terms such as “T2 长信号” to “T2 Hyperintense” and “脑实质” to “brain parenchyma”. Thirdly, leveraging this lexicon as a knowledge base, we utilized RAG techniques and prompt engineering strategies through the advanced capabilities of ChatGPT-4o,³³ to facilitate the translation of the reports from Chinese to English. Finally, the translated reports are thoroughly reviewed and refined by board-certified neuroradiologists to ensure their accuracy.

S2. Ethics approval for datasets from medical institutions

All data obtained from participating private medical institutions were obtained with ethics approval from their respective Institutional Review Boards (IRBs). This includes Xiangya Hospital, Changde First People’s Hospital, Shanghai Tongji Hospital, University of South China Second Hospital, Shenzhen Second People’s Hospital, the Third Xiangya Hospital, Jiangxi Provincial People’s Hospital, Chongqing Traditional Chinese Medicine Hospital, the First Affiliated Hospital of Nanchang University, Hunan Provincial Children’s Hospital, the First Affiliated Hospital of Lanzhou University, and the Second Affiliated Hospital of Nanchang University. The requirement for informed consent was waived by the IRBs. All data were de-identified in compliance with institutional policies and ethical standards.

S3. Estimation of total subject count from online resources

Accurately estimating subject-level statistics from the aggregated 36 online resources is challenging due to the inconsistent presence or absence of unique patient identifiers. For resources that do provide explicit patient identifiers, such as TCIA and BraTS, patient counts were directly extracted. In contrast, for web-crawled resources like PubMed Central²⁷ and ImageCLEF,²⁹ we conservatively assigned one unique patient per composite figure. This methodology likely results in an underestimate, as subfigures within composites may pertain to distinct cases, and images from the same video may originate from different patients. For certain directly downloaded datasets, including the Br35H,²¹ Brain Tumor Classification Dataset,²⁴ and Brain Tumor MRI Images 44 Classes, which lack patient identifiers, we excluded these from our subject count estimation. The detailed number of patients and their associated multi-modal data in each online resource are illustrated in Supplementary Table 2.

S4. Development of BrainVLM for tumor classification and report generation

Developing an AI model for early brain tumor diagnosis necessitates the effective integration of presurgical multi-modal data, including multi-parametric MRI scans, patient metadata, and radiology reports. Current AI-based brain diagnostic approaches largely rely on the subjective interpretation of pure MRI data, sometimes supplemented with patient metadata.^{34,35} This neglects the rich, complementary information available in multi-modal datasets, potentially limiting the model’s performance. Besides, the current vision language foundation models (VLMs³⁶) face challenges as the input length increases with the quantity and resolution of images. This is particularly alarming in brain tumor studies, which require the analysis of high-resolution (HR), multi-parametric MRI scans (T1, T1c, T2, FLAIR) from various views (axial, coronal, sagittal).

To address this, we developed BrainVLM, a first-of-its-kind VLM that effectively synergizes multi-modal medical visual and textual information. It tackles the challenges through optimization across three aspects: (1) Innovative multi-modal model architecture: Incorporating anatomical detail-aware vision encoder and hybrid causal-full attention strategies into a multi-modal network to handle complex multi-modal data efficiently. (2) Clinical-motivated data standardized: Developing a standardized dataset processing protocol that emulates clinical practices and incorporates special markers to differentiate different visual data. (3) Progressive three-stage training strategy: Using a progressive training approach to incrementally enhance the model’s multi-modal processing capabilities.

S4.1. Multi-modal model architecture.

As shown in Fig. 1b, BrainVLM consists of three core components: (1) an anatomical detail-aware vision encoder for efficient, fine-grained image feature extraction, (2) a 2-layer Multilayer Perceptron (MLP) projector that bridges the image encoder with the LLM, and (3) a hybrid causal-full attention-based LLM for reasoning and understanding the multi-modal information.

Anatomical detail-aware vision encoder. We employed the vision backbone of BioMedCLIP³⁷ as the basic visual encoder E . This model constitutes a Vision Transformer (ViT) pre-trained on a large-scale 2D medical imaging dataset using vision-language contrastive learning.³⁸ The standard ViT of original BioMedCLIP accepts only 224×224 input dimensions and cannot directly process high-resolution (HR) MRI slices (typically 400–500 pixels in width/height). Direct downsampling risks losing critical diagnostic details such as tumor margins and signal heterogeneity, while splitting the image into several sub-images for individual encoding results in excessive visual tokens for LLMs. This is computationally prohibitive due to the transformer’s quadratic complexity. We therefore designed an anatomical detail-aware vision encoder to capture finer anatomical details while maintaining computational efficiency.

Our designed visual encoder incorporated two key components: mixed-resolution visual encoding and attention-based HR injection. Let I denote an input MRI scan with dimensions $N \times 3 \times w_o \times h_o$, where N represents the number of slices (e.g., $N = 1$ for single-slice MRI; $N = 32$ for a 3D MRI volume with 32 MRI slices), and $w_o \times h_o$ specifies the original spatial resolution. We first generated two resized versions: 1) a low-resolution (LR) input I^l scaled down to $N \times 3 \times 224 \times 224$, and 2) an HR input I^h scaled up to size $N \times 3 \times 448 \times 448$. The LR input was encoded through the visual encoder E to generate global LR tokens $z^l = E(I^l) \in \mathbb{R}^{N \times 768}$. Here, z^l is extracted from the final layer’s [CLS] token of the ViT, capturing global patterns.

Next, I^h was partitioned into four non-overlapping HR sub-inputs $\{I_k^h\}_{k=1}^4$ using a grid method,³⁹ where k indexes the HR sub-inputs. Each HR sub-input was resized to $N \times 3 \times 224 \times 224$ and independently encoded by E to generate HR visual tokens as $z^{h_k} = E(I_k^h)$. Unlike the global [CLS] token extraction used for the LR input, we retained all spatial patch tokens (excluding [CLS]) from the ViT’s final layer when processing HR sub-inputs. This yields features $z^{h_k} \in \mathbb{R}^{N \times 196 \times 768}$ that encode fine-grained anatomical details. The final combined HR representation is denoted as $z^h = \{z^{h_k}\}_{k=1}^4 \in \mathbb{R}^{4 \times N \times 196 \times 768}$.

Finally, we injected z^h into z^l via an LR-HR cross-attention module, to obtain HR-informed LR tokens z^{hl} . Here, z^l served as query Q , $\{z^{h_k}\}_{k=1}^4$ served as the context (key K and value V), resulting the final $z^{hl} \in \mathbb{R}^{N \times 768}$. This approach merged global context z^l with anatomical details from z^h , while maintaining computational efficiency.

For multi-parametric MRI based brain tumor diagnosis (e.g., T1, T1c, T2, FLAIR), each 3D MRI sequence was processed independently by our encoder to extract modality-specific features, denoted as $z_m^{hl} \in \mathbb{R}^{N \times 768}$. These features were then concatenated along the slice dimension: $z^{\text{comb}} = \parallel_{m=1}^M z_m^{hl} \in \mathbb{R}^{(M \cdot N) \times 768}$, where M denotes the number of MRI sequences. For example, a 5-sequence protocol (e.g., T1 axial, T1c axial, T1c sagittal, T2 axial, FLAIR axial) with $N = 32$ slices per sequence yielded a combined representation $z^{\text{comb}} \in \mathbb{R}^{160 \times 768}$, integrating complementary diagnostic information across sequences.

MLP projector. The combined feature representation z^{comb} was mapped into the LLM language space via a 2-layer MLP projector, converting visual information into a format digestible for LLM. This transformation bridged the modality gap by converting anatomical visual patterns into a sequential token format compatible with linguistic processing.

Hybrid causal-full attention based LLM. The MLP-projected visual embeddings were combined with patient meta-data (e.g., age and gender) and textual instructions to form a multimodal prompt. This prompt was then processed by our adapted LLM for tumor classification, uncertainty quantification, and report generation. We utilized the pre-trained Llama-3.1-8B-Instruct⁴⁰ model as our base LLM, which provides a mainstream language backbone with grouped-query attention and open-source adaptability. Traditional causal attention in Llama-3.1-8B-Instruct restricts tokens to preceding positions only, which is suboptimal for MRI interpretation.⁴¹ Therefore, we implemented a hybrid attention strategy following previous works.^{42,43} Specifically, full attention was applied to image tokens for capturing complex visual interdependencies within the same 3D MRI sequence or 2D MRI slice, while causal attention was retained for sequential tokens to preserve data structure. This hybrid architecture enabled the model to simultaneously capture cross-modal visual relationships and maintain linguistic coherence.

S4.2. Clinical-motivated data standardization protocol

Language data in LLM is typically uniform and consistent, while our task involves more complex visual data. Randomly feeding unprocessed neuroimaging inputs from different modalities and views into BrainVLM complicates the learning process and impedes model convergence. Thus, we proposed a standardized dataset processing protocol that emulates clinical practices through the following three key strategies: 1) volumetric slice-count harmonization to ensure uniform slice dimensions, 2) clinical MRI sequence integration mirroring radiological workflows, and 3) special markers to differentiate multi-modal data and task-driven prompt design to differentiate different tasks. This protocol reduced learning ambiguity while preserving diagnostically critical features essential for brain tumor analysis.

MRI preprocessing via volumetric slice-count harmonization. The 3D MRI sequences used in our study typically exhibit substantial variability in the number of acquired slices. This variation occurs both across scanners/institutions (ranging from dozens to hundreds of slices per sequence) and between sequences for the same subject (e.g., 20 slices in native T1 vs. 40 in T1c sequences). This substantial slice variation significantly complicates model training. We therefore standardized each 3D MRI sequence to 32 slices using cubic spline interpolation (implemented via SimpleITK⁴⁴), ensuring smooth volumetric resampling along the slice direction. Critically, unlike conventional neuroimaging pipelines that typically involve extensive preprocessing — such as resampling to isotropic voxels and skull stripping — BrainVLM performs *only* this volumetric slice-count harmonization. This minimal preprocessing preserves the complete information content of the original MRI data, making it particularly well-suited for our comprehensive brain tumor characterization study. While conventional skull-stripping may aid certain tumor analyses, it is not universally appropriate. Crucially, lesions located in regions like the pia mater or invading the cranial vault can be inadvertently removed during skull stripping, leading to the loss of critical anatomical and pathological information — a risk our approach deliberately avoids (Supplementary Figure 5).

Clinical MRI sequence integration. Emulating the clinical workflow for MRI-based tumor identification is critical for aligning machine learning models with diagnostic practice, thereby improving lesion identification and characterization.^{45,46} In clinical practice, comprehensive brain tumor assessment often initiates with acquiring axial T1 and T2 sequences, *optionally* complemented by axial FLAIR imaging. Following intravenous administration of gadolinium-based contrast agents, multiple-plane T1c (axial, sagittal, and coronal views) are obtained to evaluate enhancement patterns and blood-brain barrier disruption. Based on initial findings or clinical suspicion, additional sequences or planes (e.g., coronal or sagittal T1/T2 for sellar region lesions) may be acquired. During interpretation, neuroradiologists typically begin by examining T2 or FLAIR images to identify potential lesions. They then use T1c images to detect small lesions and assess tumor enhancement, comparing these with T1 images of the same view for confirmation.

To emulate this clinical workflow and streamline model learning, we organized the MRI data to reflect both the clinical acquisition and interpretation workflows. Specifically, our integration strategy prioritizes clinically meaningful comparisons: the primary input consisted of paired native T1 and T1c scans acquired in the same plane, facilitating direct assessment of lesion enhancement. This core pairing was supplemented by representative T2 and FLAIR scans, as well as an additional T1c scan obtained from a different imaging plane than the primary T1/T1c pair. For example,

if the primary T1/T1c pair was axial, we included an additional T1c scan from the coronal or sagittal plane to provide multiplanar context. When specific sequences or planes were unavailable, alternative available sequences were included to maximize diagnostic coverage. In summary, BrainVLM utilized five core 3D MRI sequences as visual input—T1 (axial or another view), T1c from the same view as T1, T2 (axial or another view), FLAIR (axial or another view), and an additional T1c from a different view than T1. This structured yet flexible approach maintained comprehensive diagnostic information while faithfully reflecting real-world clinical reasoning. Further details regarding the processing of cases with more than five MRI sequences are provided in the following Model Inference Section S6.

Special tokens for data type differentiation and task-driven prompt design. BrainVLM processed heterogeneous inputs spanning 2D MRI slices, 3D MRI volumes, and free-text patient metadata (e.g., age, sex), while concurrently addressing distinct task types including classification, report generation, and reliability assessment. To enable the model to distinguish between these data types and tasks, we implemented the following specialized tokens and task-driven prompts. Specifically, to help the model distinguish different data types, we designed special tokens: 2D single image slices were enclosed with '<image>' and '</image>', while 3D MRI volumes used '<vid>' and '</vid>' tags. A special token '<s>' was inserted between volumes to indicate the different sequence's input. We also introduced specialized tumor tokens (e.g., '<class_0>' for MET) to encode the original tumor names for classification. Concurrently, to enhance the model's task-specific learning capabilities, we designed customized prompts that explicitly inform the model of the input structure and expected objectives (see Supplementary Table 7). Specifically, the finalized prompt that was input into BrainVLM integrates three components: (1) predefined special visual tokens (e.g., <vid>, <image>) to structure the imaging inputs; (2) patient metadata; and (3) task-specific instructions that explicitly define the model's objective for each task (e.g., generating a clinical description for patients in the radiology report generation task or making a self-assessment confidence score for the reliability task). These special tokens and unified prompt design ensure robust alignment between input interpretation and clinical task requirements, enhancing the model's interpretative accuracy and adaptability to diverse diagnostic objectives.

S4.3. Three-stage progressive training strategy of BrainVLM

Our curated BrainTumor46K dataset comprises three distinct data types: 53.14% 2D MRI slices paired with text descriptions, 38.14% multi-parametric 3D MRI volumes with radiology reports, and 8.72% 3D MRI sequences annotated solely with diagnostic labels. Direct end-to-end training on this heterogeneous data composition may overwhelm models with conflicting objectives. The 2D isolated MRI slices lack volumetric context essential for tumor characterization, while complete volumetric inputs demand foundational slice-level recognition capabilities that 2D training establishes. To address these challenges, we implemented a curriculum learning paradigm with three progressive training stages. This strategy enables the model to progressively interpret complex visual patterns, starting from simple 2D slices analysis and advancing to more intricate 3D volumetric data interpretation (see Supplementary Figure 3).

Training stage 1: 2D MRI slice-text representation learning. Volumetric tumor analysis fundamentally requires slice-level recognition capabilities, a skill radiologists cultivate through detailed examination of individual MRI slices. To establish this foundation while maximizing BrainTumor46K's heterogeneous resources, we initiated BrainVLM's training with a focus on 2D MRI slice-level representation learning. During this phase, we optimized only the MLP projector and the LR-HR cross-attention module within our anatomical detail-aware encoder, keeping the vision encoder and LLM module frozen to maintain stability. This stage focused on training 2D MRI slice-text alignment and utilized two main sources: (i) all available 2D MRI slice-text pairs in BrainTumor46K, and (ii) strategically extracted tumor slices from 3D MRI volumetric sequences, paired with AI-generated slice-level text descriptions. For public datasets with tumor masks, such as BraTS, we extracted slices with more than 5% tumor occupancy and generated clinically validated text descriptions using RAG, refined by ChatGPT-4.0. For institutional scans, neuroradiologists manually selected pathologically significant slices from each 3D MRI sequence, and the corresponding slice-level reports were crafted using RAG refined by ChatGPT-4.0. Finally, this meticulous process resulted in 10,996,702 curated 2D MRI slice-text pairs. This 2D MRI slice-level representation learning approach enabled BrainVLM to establish fundamental tumor recognition capabilities, laying a solid foundation for subsequent advanced 3D volumetric analysis.

Training stage 2: Volumetric context integration via hybrid 2D-3D training. In this phase, we aimed to transfer the learned 2D representation capabilities into the tomographic 3D MRI data space, facilitating BrainVLM's 3D tumor characterization capability. We utilized all available 2D and 3D training datasets from BrainTumor46K and designed a dynamic learning strategy interleaving 2D slice-level and 3D volumetric tasks during training. Crucially, at each iteration, we stochastically sampled training instances with 30% probability allocated to 2D tasks (leveraging isolated

slices for tumor description and slice-level diagnosis) and 70% probability to 3D tasks (utilizing full sequences for volumetric report generation, classification, and diagnosis). During this stage, the MLP projector, LR-HR cross-attention module, and Low-Rank Adaptation (LoRA)⁴⁷ parameters in the LLM model were trained, while the vision encoder and other parameters in the LLM model remained frozen. This transitional stage ensures that the model develops a cohesive understanding of tumor characteristics across different dimensions.

Training stage 3: Comprehensive 3D volumetric instruction tuning. In the final stage, we focused on leveraging the full high-quality 3D data available in the BrainTumor46K dataset to enhance BrainVLM’s advanced volumetric analysis capabilities. The projector, LR-HR cross-attention module and LoRA parameters in the LLM model were trained, while all other parameters remained frozen.

S4.4. Loss function

We introduced dual-objective losses to unleash BrainVLM’s generative and discriminative capabilities across clinical tasks, including free-text generation and dedicated tumor classification. Specifically, we first employed a standard causal language modeling loss $\mathcal{L}_{\text{LM}}$, which enables the model to follow natural language instructions and generate coherent diagnostic texts. Given a token sequence $(w_i^{(1)}, w_i^{(2)}, \dots, w_i^{(T)})$ in a training sample x_i , this loss is formulated as:

$$\mathcal{L}_{\text{LM}} = - \sum_{t=1}^T \log P(w_i^{(t)} | w_i^{(1)}, w_i^{(2)}, \dots, w_i^{(t-1)}; \theta),$$

where $P(w_i^{(t)} | w_i^{(1)}, w_i^{(2)}, \dots, w_i^{(t-1)}; \theta)$ denotes the predicted probability of token w_t given the preceding tokens. θ denotes the model parameters. $t = 1, \dots, T$ denotes the tokens in the sequence. This objective encourages contextual understanding and fluent text generation by minimizing the negative log-likelihood of ground-truth tokens.

Furthermore, to enhance the tumor-specific discrimination, we introduced an auxiliary classification loss $\mathcal{L}_{\text{CL}}$ to perceive fine-grained medical patterns. We added special tokens to encode the original tumor label (such as <tumor_3> token for Glioma), and employed a classification loss based on these tokens. This loss is computed as:

$$\mathcal{L}_{\text{CL}} = - \sum_{c=1}^{14} y_c^{(i)} \log \hat{y}_c^{(i)},$$

where $y_c^{(i)}$ denotes the ground-truth label for the c -th tumor class, and $\hat{y}_c^{(i)}$ is the predicted probability. This loss enables fine-grained recognition across 14 tumor categories, including 11 major brain tumor types in WHO-CNS5 and 3 GNN subcategories.

The overall objective combines both losses:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{LM}} + \alpha \mathcal{L}_{\text{CL}},$$

where α is a tunable coefficient balancing language generation and tumor classification. This joint optimization framework enables BrainVLM to effectively capture both general linguistic semantics and task-specific diagnostic cues.

S4.5. Data augmentation

Brain MRI data exhibited domain shifts driven by patient-specific factors (e.g., age, gender, physique) and MRI hardware variability,⁴⁸ resulting in significant intra-modality variation. We applied tailored augmentation strategies for both 2D and 3D MRI data to improve BrainVLM’s generalization across diverse clinical settings. For 2D MRI slices, we adopted standard augmentation techniques,⁴⁹ including random flipping, cropping, and intensity transformations. For 3D MRI sequences, we first identified key challenges such as noise artifacts, intensity inhomogeneity, and motion-induced blurring (examples shown in Supplementary Figure 4). Then, to improve robustness against these effects, each 3D training sample was augmented with: (1) random rotation within ± 60 degrees, and (2) intensity variation, motion blur, or random noise, each applied with 30% probability.

S5. Model reliability implement

Brain tumor diagnosis is a high-stakes task, where a misdiagnosis can lead to inappropriate treatments, prolonged patient suffering, and potentially fatal outcomes. However, conventional VLMs may generate erroneous predictions

with high confidence,⁵⁰ undermining clinical trust and limiting practical utility. As illustrated in the left panel of Fig. 2c in the main manuscript, our vanilla BrainVLM model (without reliability mechanisms) also exhibited this concerning behavior, frequently assigning high confidence scores to erroneous predictions. To address this critical limitation, we presented an uncertainty quantification strategy that calibrates prediction confidence in BrainVLM: accurate diagnoses receive low uncertainty scores (high confidence score), while erroneous predictions are assigned high uncertainty (low confidence score). This ensures only high-confidence, reliable predictions are acted upon.

Existing uncertainty quantification approaches for VLMs and LLMs rely on multiple sampling strategies (e.g., Best-of-N) that incur substantial computational overhead regardless of question complexity.⁵¹ While adaptive sampling techniques attempt to reduce costs through early-stopping heuristics, they remain constrained by manually designed rules that lack generalizability across models and tasks.⁵² We introduce a consensus-driven confidence calibration strategy that directly transforms model disagreement into calibrated confidence signals, eliminating the need for heuristic rules or fixed sampling costs. Our approach features two key innovations: (1) Consensus-driven reliability dataset construction, where a reliability dataset is automatically constructed using agreement statistics from multiple model instances via consensus-driven clustering; (2) Confidence-aware reliable BrainVLM finetuning: Distillation of consensus-derived reliability knowledge into our BrainVLM framework via parameter-efficient fine-tuning. We refer to the baseline BrainVLM without uncertainty quantification as the vanilla BrainVLM, while our enhanced version incorporating the proposed uncertainty quantification strategy is termed reliable BrainVLM.

S5.1. Consensus-driven reliability dataset construction

To create a high-quality reliability dataset for confidence calibration, we leveraged the natural training dynamics of BrainVLM. Specifically, when our best-performing model was near convergence, we sampled N model checkpoints at different training iterations to create an ensemble of N vanilla BrainVLM variants. For each training sample x_i with pathological label $y_{\text{true}}^{(i)}$, we independently processed x_i through all N models to obtain N diagnostic predictions $\{\hat{y}_i^{(1)}, \dots, \hat{y}_i^{(N)}\}$ per sample (see Supplementary Figure 3a). These responses were then clustered by predicted diagnoses, with the largest cluster defining the consensus label $y_{\text{cons}}^{(i)}$. Formally, the consensus label is computed as:

$$y_{\text{cons}}^{(i)} = \arg \max_{c \in \mathcal{C}} \sum_{k=1}^N \mathbb{I}(\hat{y}_i^{(k)} = c),$$

where $\mathcal{C}$ denotes the set of all possible diagnostic classes and $\mathbb{I}$ indicates the indicator function. This consensus label represents the prediction with the strongest inter-model agreement. We then computed a confidence score $s_{\text{con}}^{(i)}$ for each sample x_i :

$$s_{\text{con}}^{(i)} = \frac{N - N_w^{(i)}}{N},$$

where $N_w^{(i)} = \sum_{k=1}^N \mathbb{I}(\hat{y}_i^{(k)} \neq y_{\text{true}}^{(i)})$, which indicates the number of incorrect predictions among the N models. A high N_w implies greater disagreement, resulting in a lower confidence score, while lower N_w yields higher confidence scores.

For each sample $(x_i, y_{\text{true}}^{(i)})$, the above process yields per-sample supervision tuples $(x_i, y_{\text{cons}}^{(i)}, s_{\text{con}}^{(i)})$, where $y_{\text{cons}}^{(i)}$ represents the ensemble-aggregated prediction, and $s_{\text{con}}^{(i)}$ quantifies reliability based on inter-model consensus strength $y_{\text{cons}}^{(i)}$ and clinical validity alignment with $y_{\text{true}}^{(i)}$. Collectively, these tuples constitute the reliability dataset, which provides reliable supervision for training BrainVLM to generate well-calibrated, confidence-aware predictions.

S5.2. Confidence-aware reliable BrainVLM finetuning

After the construction of the reliability dataset, we then distilled the consensus-derived reliability knowledge into our reliable BrainVLM through confidence-aware reliable fine-tuning. Specifically, as illustrated in Supplementary Figure 3b, we transformed the vanilla BrainVLM into reliable BrainVLM by parameter-efficient fine-tuning on our constructed reliability dataset. In this process, we also reformulated the output scheme to generate an additional confidence score, yielding responses such as: “Report:..., diagnosis: <tumor_i>. My confidence: 80%”. Crucially, we employed our curated tuples $(x_i, y_{\text{cons}}^{(i)}, s_{\text{con}}^{(i)})$ rather than ground-truth pairs $(x_i, y_{\text{true}}^{(i)})$ as supervision signals, enabling simultaneous learning of diagnostic predictions and their associated confidence levels. During this fine-tuning process, we updated only the projector, LR-HR cross-attention module, and LoRA parameters within the LLaMA model,

340 keeping both the vision encoder and LLaMA backbone frozen. The model was trained for two epochs on the reliability-annotated dataset.

S6. Model inference

Let $\mathcal{S} = \{\text{T1}, \text{T1c}, \text{T2}, \text{FLAIR}\}$ denote the set of MRI sequences. For each sequence $s \in \mathcal{S}$, let $\mathcal{V}_s \subseteq \{\text{axial}, \text{sagittal}, \text{coronal}\}$ represent the available views for s ; $\mathcal{V}_s$ may be empty if the corresponding sequence is missing.

345 Given a patient, the complete input MRI space can be formulated as

$$\mathcal{X} = \{(x_s^v, s, v) \mid s \in \mathcal{S}, v \in \mathcal{V}_s\}, \quad (1)$$

where x_s^v denotes the MRI image corresponding to sequence s and view v . As described in Clinical MRI sequence integration Section S4.2), during the inference time, BrainVLM takes five core MRI sequences—T1 (axial or another view), T1c from the same view as T1, T2, FLAIR, and an additional T1c from a different view than T1—along with patient metadata and task-specific instructions as input. Formally, the five core MRI sequences C_m satisfy:

$$C_m = \{(x_{\text{T1}}^{v_0}, \text{T1}, v_0), (x_{\text{T1c}}^{v_0}, \text{T1c}, v_0), (x_{s_3}^{v_3}, s_3, v_3), (x_{s_4}^{v_4}, s_4, v_4), (x_{s_5}^{v_5}, s_5, v_5)\} \quad (2)$$

350 where $v_0, v_3, v_4, v_5 \in \mathcal{V}_s$ are chosen from available views, and the selection follows these criteria:

- $v_0 \in \mathcal{V}_{\text{T1}}$ is any available view of T1 (e.g., axial, sagittal, or coronal).
- The first two items are T1 and T1c from the same view v_0 .
- For s_3 and s_4 , select from $\{\text{T2}, \text{FLAIR}\}$ according to availability:
 - If both T2 and FLAIR are available, then $s_3 = \text{T2}$, $s_4 = \text{FLAIR}$, with $v_3 \in \mathcal{V}_{\text{T2}}$, $v_4 \in \mathcal{V}_{\text{FLAIR}}$.
 - 355 – If only one of T2 or FLAIR is available, assign it to s_3 and select any other available MRI sequence for s_4 .
 - If neither T2 nor FLAIR is available, select any other available MRI sequences for s_3 and s_4 .
- For s_5 , prioritize selecting T1c from any view $v_5 \in \mathcal{V}_{\text{T1c}}$ such that $v_5 \neq v_0$. If not available, select any unused view from the remaining available sequences.

360 This selection strategy dynamically adapts to the available sequences and views for each patient, ensuring robust performance even in the presence of missing sequences. During inference, BrainVLM takes the selected five core MRI sequences, along with patient metadata and task-specific instructions, as input. It then generates a diagnostic prediction, a radiology report, and an associated confidence score.

365 **Adaptive sequence sampling and aggregation.** In real-world clinical practice, MRI studies frequently employ thick-slice acquisitions spanning multiple anatomical planes (e.g., axial, sagittal, and coronal views for both T1 and T2 sequences), yielding studies exceeding the standard five-sequence input. To address this variability, we define an adaptive sequence sampling strategy as follows.

Let $\mathcal{C}$ be the set of all valid five-sequence combinations constructed from $\mathcal{X}$ according to the above selection criteria:

$$\mathcal{C} = \{C_m \subset \mathcal{X} \mid |C_m| = 5 \text{ and } C_m \text{ satisfies the selection constraints above}\}. \quad (3)$$

For each $C_m \in \mathcal{C}$, we define the model inference function:

$$(y_m, r_m, s_{\text{con}}^m) = f_\theta(C_m, d, t), \quad (4)$$

370 where f_θ defines the reliable BrainVLM parameters, d denotes patient metadata, t denotes the task-specific prompt, y_m is the predicted clinical diagnosis, r_m is the generated radiology report, and s_{con}^m is the confidence score. To aggregate results across all sampled combinations, we define the final clinical diagnosis $\hat{y}^{(1)}$ as the most frequent prediction (“majority voting”) among all $\{y_m\}_{C_m \in \mathcal{C}}$:

$$\hat{y}^{(1)} = \text{mode}(\{y_m\}_{m=1}^M) \quad (5)$$

where $\text{mode}(\cdot)$ indicates the statistical mode (most frequent element) in the ensemble. Here $\hat{y}^{(1)}$ is also known as the TOP 1 prediction. The final confidence $\hat{s}_{con}^{(1)}$ that associated with the predicted diagnosis $\hat{y}^{(1)}$ can be aggregated as:

$$\hat{s}_{con}^{(1)} = \frac{1}{N_1} \sum_{m=1}^M s_{con}^m \cdot \delta(y_m, \hat{y}_1), \quad N_1 = \sum_{m=1}^M \delta(y_m, \hat{y}_1), \quad (6)$$

where $\delta(a, b) = 1$ if $a = b$, else 0. The final radiology report is generated by aggregating reports associated with the TOP 1 diagnosis. Specifically, since our report are structured into predefined sections (e.g., T1 signal characteristics), we apply majority voting within each section to select the most frequent content among all reports associated with the TOP 1 predicted diagnosis $r_m \mid y_m = \hat{y}^{(1)}$.

This adaptive ensemble strategy ensures that the model can robustly process heterogeneous and incomplete clinical MRI acquisitions by (1) systematically sampling all valid five-sequence input sets from the available MRI space $\mathcal{X}$, (2) independently inferring on each combination, and (3) performing robust aggregation via majority voting to determine the final (Top 1) clinical prediction.

S6.1. Confidence-triggered TOP 2 supplementary diagnosis.

In clinical workflows, neuroradiologists may consider multiple diagnostic possibilities when interpreting clinically ambiguous cases, such as “considering meningioma or glioma”. To emulate this process, we implement a confidence-triggered supplementary diagnosis mechanism. If the confidence score $\hat{s}_{con}^{(1)}$ for the Top 1 prediction $\hat{y}^{(1)}$ falls below a predefined threshold (set at 75% in this study), a secondary (top-2) diagnosis is also reported. The Top-2 diagnosis $\hat{y}^{(2)}$ is defined as: $\hat{y}^{(2)} = \text{mode}(\{y_m \mid y_m \neq \hat{y}_1\})$. Our experimental results in Figure 1c, Figure 3d and Figure 3e in the main manuscript demonstrate the effectiveness of this approach in enhancing diagnostic reliability in ambiguous cases.

S7. Finetuning of molecular subgroup classification task

In this phase, we fine-tuned BrainVLM using the stage-3 checkpoint, leveraging the adult-type diffuse glioma data from BrainTumor46K. We performed threefold validation on the Xiangya primary Dataset (n = 494; 333 IDH-w GBMs, 86 IDH-m Astros, 75 IDH-m ODGs), each fold consisted of 328 patients (222 IDH-w GBMs, 57 IDH-m Astros, 50 IDH-m ODGs) for training and 166 patients (111 IDH-w GBMs, 29 IDH-m Astros, 25 IDH-m ODGs) for validation. We followed the MRI sequences integration strategy defined in the Section S4.2 and S6 for molecular classification training and prediction. In this training stage, we updated only the projector, LR-HR cross-attention module and LoRA parameters within the LLaMA model, keeping other parameters frozen.

The results demonstrated BrainVLM’s robust capability to differentiate molecular subtypes of adult diffuse gliomas. BrainVLM achieved a frequency-weighted F_1 score of 0.92 (95% CI: 0.91–0.93, Supplementary Figure 7c) in the threefold validation on the Xiangya primary dataset, outperforming Merlin ($F_1 = 0.76$, 95% CI: 0.75–0.77) and RadFM ($F_1 = 0.74$, 95% CI: 0.73–0.75). Across molecular subgroups, BrainVLM demonstrated superior performance: IDH-w GBMs ($F_1 = 0.95$ vs. 0.88–0.91), IDH-m Astros ($F_1 = 0.89$ vs. 0.53–0.62), and IDH-m ODGs ($F_1 = 0.84$ vs. 0.58–0.63). On external validation, BrainVLM achieved a frequency-weighted F_1 score of 0.88 (Supplementary Figure 7c), surpassing Merlin ($F_1 = 0.73$) and RadFM ($F_1 = 0.69$). Consistent with primary validation, BrainVLM showed higher performance across subgroups: IDH-w GBMs ($F_1 = 0.94$ vs. 0.82–0.85), IDH-m Astros ($F_1 = 0.79$ vs. 0.32–0.44), and IDH-m ODGs ($F_1 = 0.69$ vs. 0.27–0.34).

S8. Implementation details

We employed AdamW⁵³ with weight decay regularization ($\lambda=0.05$) across all training stages, coupled with a cosine decay scheduler and 300-step linear warmup. During the training stage 1 (2D MRI-text representation learning), the learning rate was linearly scaled from 1×10^{-6} to 1×10^{-4} , and the batch size was set to 256. For training stage 2 (hybrid 2D-3D training) and stage 3 (3D volumetric learning), the learning rate followed the same warmup initialization (1×10^{-6}). However, the peak learning rate was set to 3×10^{-5} for training stage 2 and 1×10^{-5} for training stage 3, respectively. For the molecular classification finetuning, the peak learning rate was set to 1×10^{-4} . We set the batch size to 1 on each GPU device for both training stages 2 and 3, resulting in a total effective batch size of 4. All experiments and the implementation of the BrainVLM were conducted using Python (version 3.9). We also utilized

torch (2.0.0), transformers (4.45.2), huggingface-hub (0.29.2), tokenizers (0.20.3) and peft (0.2.0) for model training. For image processing and data augmentation, scikit-image (0.24.0), SimpleITK (2.4.1), monai (1.4.0), numpy (1.26.4),
 420 opencv-python (4.7.0) and nibabel (5.3.2) were applied. For training stage 1, we used 8 NVIDIA 4090 GPUs for multi-GPU training. For the training stages 2 and 3, as well as the reliable training, we used 4 NVIDIA L40 GPUs.

S8.1. Competing methods

We compared BrainVLM with three state-of-the-art VLMs: RadFM,⁵⁴ Merlin,⁵⁵ and ChatGPT-4o.³³ RadFM is a radiology foundation model pretrained on 15.5 million 2D medical images and 500,000 3D medical scans. RadFM
 425 utilizes a 12-layer 3D ViT as a vision encoder to process both 2D and 3D data, and uses MedLLaMA-13B as the large language model for reasoning. For our experiments, we fine-tuned the pretrained RadFM⁵⁴ on our BrainTumor46K dataset, updating the LoRA parameters of MedLLaMA-13B and the projection module. Merlin is a 3D vision-language foundation model developed for abdominal CT interpretation. It was pretrained on both radiology reports and structured electronic health records (EHRs), using a 3D ResNet152 as the vision encoder and RadLlama-7B
 430 as the language model.⁵⁵ The vision encoder was pretrained on a large-scale paired dataset of 3D CT scans and text reports using the InfoNCE loss. For our experiments, we fine-tuned Merlin’s vision encoder based on its pretrained 3D ResNet152 weights, and applied LoRA-based fine-tuning to the RadLlama-7B language model. Both RadFM and Merlin are designed for single-modality radiology inputs. For a fair comparison, we adopted the same input strategy in BrainVLM as in these baseline models. Specifically, we provided five MRI sequences as input, processed each
 435 sequence through the respective vision encoders, and concatenated the extracted features before passing them to the language model. To ensure a rigorous and fair comparison, we extensively fine-tuned all baseline models on the BrainTumor46K dataset with sufficient hyperparameter optimization, training each model until convergence.

ChatGPT-4o³³ does not natively support 3D medical imaging input. Therefore, for each test case, experienced clinicians manually selected the most representative tumor-containing slice from each MRI sequence (four to five
 440 sequences per case). These clinician-validated slices were combined and provided as a set of 2D images to ChatGPT-4o for report generation and diagnostic prediction. It is important to note that this approach simplifies the diagnostic task, as tumor-containing frames are pre-identified by experts, whereas BrainVLM, RadFM, and Merlin operate directly on the full 3D raw MRI volumes without manual selection.

S9. Quantitative assessment and statistical analysis

445 We evaluated BrainVLM’s tumor prediction performance using sensitivity, specificity, precision, Cohen’s *Kappa*, F_1 , and Area Under the Curve (AUC). We used both the common language metrics (Bilingual Evaluation Understudy (BLEU)⁵⁶) and clinical relevance metrics (Ratescore⁵⁷ and RadGraph-XL⁵⁸) to measure the BrainVLM’s report generation performance. Given true-positive (TP), false-positive (FP), true-negative (TN) and false-negative (FN) rates, the precision, sensitivity (recall), specificity, and F_1 score of each class are calculated as follows:

$$Precision = \frac{TP}{TP + FP},$$

$$Sensitivity = \frac{TP}{TP + FN},$$

$$Specificity = \frac{TN}{TN + FP},$$

$$F_1\text{-score} = \frac{2 \cdot Precision \cdot Sensitivity}{Precision + Sensitivity},$$

$$Cohen's\ Kappa = \frac{P_o - P_e}{1 - P_e},$$

where

$$P_o = \frac{TP + TN}{TP + FP + TN + FN} \text{ and } P_e = \frac{(TP + FP) \times (TP + FN)}{(TP + FP + TN + FN)^2} + \frac{(TN + FN) \times (TN + FP)}{(TP + FP + TN + FN)^2}.$$

To assess the BrainVLM’s overall multi-label brain tumor classification capability, we computed the class frequency-weighted mean for both the F_1 score and the AUC. More specifically, given the F_{1_i} score for class i , the frequency-weighted F_1 score across total C classes are defined as:

$$\text{Weighted } F_1 = \sum_{i=1}^C \frac{n_i}{N} \times F_{1_i}.$$

where n_i is the number of patients in class i , N is the total number of patients across all classes.

BLEU-4 is a widely used metric for evaluating lexical similarity between machine-generated and human reference texts.⁵⁶ Unlike traditional natural language metrics, both RadGraph-XL⁵⁸ and Ratescore⁵⁷ prioritize domain-specific content critical to radiology, making them particularly suitable for evaluating automated report generation systems. RaTEScore⁵⁷ evaluates clinical reports by identifying medical entities across imaging modalities and regions, and then calculates a clinical relevance-weighted similarity score. Similarly, RadGraph-XL⁵⁸ is a metric designed to evaluate the quality of radiology reports by assessing their semantic and clinical accuracy. It constructs a graph-based representation of entities and relationships within a report, capturing anatomical findings, observations, and their interconnections, such as whether a finding is present, absent, or uncertain. By comparing these graphs between generated and reference reports, RadGraph-XL quantifies similarity using an F_1 score, providing a robust measure of how well a generated report conveys clinically relevant information.

S10. Robustness of BrainVLM to missing MRI sequences

BrainVLM performs inference using four standard MRI sequences (T1, T1c, T2, and FLAIR). In clinical practice, incomplete acquisition of MRI sequences (e.g., missing an entire sequence) frequently occurs due to insufficient clinician awareness or contraindications. To evaluate the model’s robustness to such scenarios, we conducted an ablation study by systematically excluding one of the four standard sequences. For each excluded sequence, we maintained the required five-image input structure by randomly selecting an additional view from the remaining sequences. For example, when removing T1, we substituted an extra view from T1c, T2, or FLAIR (e.g., supplementary sagittal T1c). For fair comparison, our analysis included only patients meeting the five-image requirement after sequence removal. This resulted in 2,701 patients in the primary test dataset and 870 patients in the external test dataset. As shown in Supplementary Table 8 and Supplementary Figure 7 (central panel), BrainVLM achieves diagnostic performance comparable to the full-sequence baseline in scenarios missing T2, FLAIR, or T1 sequences, with less than a 5% overall reduction in F_1 score. However, the unavailability of T1c sequences caused a 13.4% F_1 -score decline, underscoring its critical role in brain tumor assessment.

Furthermore, we employed Shapley value analysis⁵⁹ to evaluate the contribution of each input sequence (T1, T1c, T2, and FLAIR) to the model’s decisions. As shown in Supplementary Figure 7, the analysis revealed consistent patterns with our ablation results: omission of the T1c sequence led to the most substantial decline in diagnostic performance across all tumor types, highlighting its critical importance. Additionally, EMB classification exhibited uniform sensitivity to the removal of any single sequence (approximately 35 % mean performance drop), while GCN and MEN predictions remained stable when T2 or FLAIR was missing. In contrast, CPT predictions were more dependent on T1, T1c and FLAIR, but showed resilience to the exclusion of T2. These findings align well with clinical diagnostic experience, where specific sequences are known to be more informative for certain tumor types.

S11. Data availability

The online training data are available on their corresponding website. We will publish the scripts for pre-processing the online dataset upon publication of this manuscript. The institutional training data that support the findings of this study are available from the corresponding author upon request. The detailed publicly available resources can be accessed at: PubMed Central (<https://pmc.ncbi.nlm.nih.gov/>), Ctisus (<https://www.ctisus.com/>), ImageCLEFmedical23 (<https://www.imageclef.org/2023/medical/caption>), Br35H (<https://www.kaggle.com/datasets/ahmedhamada0/brain-tumor-detection>), Figshare Brain Tumor Dataset (<https://www.kaggle.com/datasets/ashkhagan/figshare-brain-tumor-dataset>), Brain Tumor Classification (<https://www.kaggle.com/datasets/sartajbhuvaji/brain-tumor-classification-mri>), Radiopaedia (<https://radiopaedia.org/home>), Brain Tumor MRI8 images 44

classes (<https://www.kaggle.com/datasets/fernando2rad/brain-tumor-mri-images-44c>), LGG-1p19qDeletion (<https://www.cancerimagingarchive.net/collection/lgg-1p19qdeletion/>), BraTS23 (<https://synapse.org/brats>), ReMIND (<https://www.cancerimagingarchive.net/collection/remind/>), RHUH-GBM (<https://www.cancerimagingarchive.net/collection/rhuh-gbm/>), UPENN-GBM (<https://www.cancerimagingarchive.net/collection/upenn-gbm/>), GLIS-RT (<https://www.cancerimagingarchive.net/collection/glis-rt/>), QIN GBM Treatment Response (<https://www.cancerimagingarchive.net/collection/qin-gbm-treatment-response/>), LUMIERE (<https://github.com/ysuter/gbm-data-longitudinal>), OpenNeuro Diffuse Gliomas (<https://openneuro.org/datasets/ds004717/versions/1.0.0>), Brain-Tumor- Progression (<https://www.cancerimagingarchive.net/collection/brain-tumor-progression/>), RIDER (<https://www.cancerimagingarchive.net/collection/rider-lung-ct/>), ACRIN-DSC- MR-Brain (<https://www.cancerimagingarchive.net/collection/acrin-dsc-mr-brain/>), ACRIN-FMISO- Brain (<https://www.cancerimagingarchive.net/collection/acrin-fmiso-brain/>), Meningioma- SEG-CLASS (<https://www.cancerimagingarchive.net/collection/meningioma-seg-class/>), Brain metastase MRI dataset (<https://www.nature.com/articles/s41597-023-02123-0>), Brain-TR- GammaKnife, AOMIC-ID1000 (<https://openneuro.org/datasets/ds003097>), AOMIC- PIOP1 (<https://openneuro.org/datasets/ds002785>), IXI (<https://brain-development.org/ixi-dataset/>), OpenNeuro Listening Task (<https://openneuro.org/datasets/ds004285/versions/1.0.0>), QTAB dataset (<https://openneuro.org/datasets/ds004146/versions/1.0.4>), OpenNeuro Visual Audiovisual Speech (<https://openneuro.org/datasets/ds003717/versions/1.0.1>), OpenNeuro Dynamic Passive Threat (<https://openneuro.org/datasets/>), OpenNeuro UCLA Consortium (<https://openneuro.org/datasets/ds000030/versions/00016>), OpenNeuro NIMH Healthy (<https://openneuro.org/datasets/ds004215>), DLBS dataset (<https://openneuro.org/datasets/ds004856/versions/1.0.0>), OpenNeuro Healthy Adults (<https://openneuro.org/datasets/ds002330/versions/1.1.0/1.1.0>). Note that some online resources (e.g., Radiopaedia) cannot be directly accessed for AI research purposes. We have obtained official authorization for non-commercial use in this study.

Code availability

BrainVLM will be fully available at <https://github.com/HKU-HealthAI/BrainVLM> upon acceptance of the paper. We will release all model weights and relevant source code for pretraining, fine-tuning, and inference to facilitate research transparency and community collaboration.

References

- 1 Baid U, et al. The RSNA-ASNR-MICCAI BraTS 2021 Benchmark on Brain Tumor Segmentation and Radiogenomic Classification. arXiv preprint. 2021:arXiv:2107.02314.
- 2 Adewole M, Rudie JD, Gbdamosi A, Toyobo O, Raymond C, Zhang D, et al. The brain tumor segmentation (BraTS) challenge 2023: glioma segmentation in sub-Saharan Africa patient population (BraTS-Africa). arXiv preprint. 2023:arXiv:2305.19369.
- 3 Kazerooni AF, Khalili N, Liu X, Haldar D, Jiang Z, Anwar SM, et al. The brain tumor segmentation (BraTS) challenge 2023: focus on pediatrics (CBTN-CONNECT-DIPGR-ASNR-MICCAI BraTS-PEDs). arXiv preprint. 2024:arXiv:2305.17033.
- 4 Moawad AW, Janas A, Baid U, Ramakrishnan D, Saluja R, Ashraf N, et al. The Brain Tumor Segmentation (BraTS-METS) Challenge 2023: brain Metastasis Segmentation on Pre-treatment MRI. arXiv preprint arXiv:230600838. 2024.
- 5 Correia de Verdier M, Saluja R, Gagnon L, LaBella D, Baid U, Hoda Tahon N, et al. The 2024 Brain Tumor Segmentation (BraTS) Challenge: glioma Segmentation on Post-treatment MRI. arXiv preprint arXiv:240518368. 2024.

- 6 LaBella D, Schumacher K, Mix M, Leu K, McBurney-Lin S, Nedelec P, et al. Brain Tumor Segmentation (BraTS) Challenge 2024: meningioma Radiotherapy Planning Automated Segmentation. arXiv preprint arXiv:240518383. 2024.
- 7 Cepeda S, et al.. The Río Hortega University Hospital glioblastoma dataset: a comprehensive collection of pre-operative, early postoperative and recurrence MRI scans (RHUH-GBM); 2023. The Cancer Imaging Archive <https://doi.org/10.7937/4545-c905>.
- 8 Pedano N, Flanders AE, Scarpance L, Mikkelsen T, Eschbacher JM, Hermes B, et al.. The Cancer Genome Atlas Low Grade Glioma Collection (TCGA-LGG) (Version 3); 2016. The Cancer Imaging Archive <https://doi.org/10.7937/K9/TCIA.2016.L4LTD3TK>.
- 9 Scarpance L, Mikkelsen T, Cha S, Rao S, Tekchandani S, Gutman D, et al.. The Cancer Genome Atlas Glioblastoma Multiforme Collection (TCGA-GBM) (Version 5); 2016. The Cancer Imaging Archive <https://doi.org/10.7937/K9/TCIA.2016.RNYFUYE9>.
- 10 Barboriak D. Data From RIDER NEURO MRI; 2015. The Cancer Imaging Archive <http://doi.org/10.7937/K9/TCIA.2015.VOSN3HN1>.
- 11 Vassantachart A, Cao Y, Shen Z, Cheng K, Gribble M, Ye JC, et al.. Segmentation and Classification of Grade I and II Meningiomas from Magnetic Resonance Imaging: an Open Annotated Dataset (Meningioma-SEG-CLASS) (Version 1); 2023. The Cancer Imaging Archive <https://doi.org/10.7937/0TKV-1A36>.
- 12 Wang Y, Duggar WN, Caballero DM, Vengaloor Thomas T, Adari N, Mundra EK, et al.. Brain Tumor Recurrence Prediction after Gamma Knife Radiotherapy from MRI and Related DICOM-RT: an Open Annotated Dataset and Baseline Algorithm (Brain-TR-GammaKnife); 2023. The Cancer Imaging Archive <https://doi.org/10.7937/xb6d-py67>.
- 13 Clark K, Vendt B, Smith K, Freymann J, Kirby J, Koppel P, et al. The Cancer Imaging Archive (TCIA): maintaining and Operating a Public Information Repository. *Journal of digital imaging*. 2013;26:1045-57.
- 14 Schmainda KM, Prah M. Data from Brain-Tumor-Progression; 2018. The Cancer Imaging Archive <https://doi.org/10.7937/K9/TCIA.2018.15quzvnv>.
- 15 Kinahan P, Muzi M, Bialecki B, Herman B, Coombs L. Data from ACRIN-DSC-MR-Brain; 2019. The Cancer Imaging Archive <https://doi.org/10.7937/tcia.2019.zr1pjf4i>.
- 16 Kinahan P, Muzi M, Bialecki B, Coombs L. Data from ACRIN-FMISO-Brain (Version 2); 2018. The Cancer Imaging Archive <https://doi.org/10.7937/K9/TCIA.2018.vohlekok>.
- 17 Erickson B, Akkus Z, Sedlar J, Korfiatis P. Data from LGG-1p19qDeletion (Version 2); 2017. The Cancer Imaging Archive <https://doi.org/10.7937/K9/TCIA.2017.DWEHTZ9V>.
- 18 Juvekar P, Dorent R, Kögl F, Torio E, Barr C, Rigolo L, et al.. The Brain Resection Multimodal Imaging Database (ReMIND) (Version 1); 2023. The Cancer Imaging Archive <https://doi.org/10.7937/3RAG-D070>.
- 19 Shusharina N, Bortfeld T. Glioma Image Segmentation for Radiotherapy: RT targets, barriers to cancer spread, and organs at risk (GLIS-RT); 2021. The Cancer Imaging Archive <https://doi.org/10.7937/TCIA.T905-zQ20>.
- 20 Bakas S, Sako C, Akbari H, Bilello M, Sotiras A, Shukla G, et al.. Multi-parametric magnetic resonance imaging (mpMRI) scans for de novo Glioblastoma (GBM) patients from the University of Pennsylvania Health System (UPENN-GBM) (Version 2); 2021. The Cancer Imaging Archive <https://doi.org/10.7937/TCIA.709X-DN49>.
- 21 Hamada A. Br35H: brain Tumor Detection 2020; 2020. Kaggle Dataset <https://www.kaggle.com/datasets/ahmedhamada0/brain-tumor-detection>.
- 22 Cheng J. Brain tumor dataset; 2017. figshare <https://doi.org/10.6084/m9.figshare.1512427.v8>.

- 23 Feltrin F. Brain Tumor MRI Images 44 Classes; 2023. Kaggle Dataset <https://www.kaggle.com/datasets/fernando2rad/brain-tumor-mri-images-44c>.
- 24 Bhuvaji S, Kadam A, Bhumkar P, Dedge S, Kanchan S. Brain Tumor Classification (MRI); 2020. Kaggle Dataset <https://www.kaggle.com/dsv/1183165>.
- 25 Johns Hopkins University School of Medicine. CTisus;. <https://www.ctisus.com/>.
- 26 Radiopaedia. Radiopaedia;. <https://radiopaedia.org>.
- 27 Roberts RJ. PubMed Central: The GenBank of the published literature. The National Academy of Sciences; 2001.
- 28 Fan L, Li H, Zhuo J, Zhang Y, Wang J, Chen L, et al. The human brainnetome atlas: a new brain atlas based on connectional architecture. *Cerebral cortex*. 2016;26(8):3508-26.
- 29 Ionescu B, Müller H, Péteri R, Abacha AB, Sarrouiti M, Demner-Fushman D, et al. Overview of the ImageCLEF 2021: Multimedia retrieval in medical, nature, internet and social media applications. In: *Experimental IR Meets Multilinguality, Multimodality, and Interaction: 12th International Conference of the CLEF Association, CLEF 2021, Virtual Event, September 21–24, 2021, Proceedings 12*. Springer; 2021. p. 345-70.
- 30 Ren S, He K, Girshick R, Sun J. Faster R-CNN: towards Real-Time Object Detection with Region Proposal Networks. In: Cortes C, Lawrence N, Lee D, Sugiyama M, Garnett R, editors. *Advances in Neural Information Processing Systems*. vol. 28. Curran Associates, Inc.; 2015. Available from: https://proceedings.neurips.cc/paper_files/paper/2015/file/14bfa6bb14875e45bba028a21ed38046-Paper.pdf.
- 31 Subramanian S, Wang LL, Mehta S, Bogin B, van Zuylen M, Parasa S, et al. MediCaT: a dataset of medical images, captions, and textual references. *arXiv preprint arXiv:201006000*. 2020.
- 32 Smith R. An overview of the Tesseract OCR engine. In: *Ninth international conference on document analysis and recognition (ICDAR 2007)*. vol. 2. IEEE; 2007. p. 629-33.
- 33 OpenAI. GPT-4; 2023. <https://openai.com/research/gpt-4>.
- 34 Wang YRJ, Wang P, Yan Z, Zhou Q, Gunturkun F, Li P, et al. Advancing presurgical non-invasive molecular subgroup prediction in medulloblastoma using artificial intelligence and MRI signatures. *Cancer Cell*. 2024;42(7):1239-57.
- 35 Havaei M, Davy A, Warde-Farley D, Biard A, Courville A, Bengio Y, et al. Brain tumor segmentation with deep neural networks. *Medical image analysis*. 2017;35:18-31.
- 36 Tong P, Brown E, Wu P, Woo S, Iyer A, Akula SC, et al. Cambrian-1: a Fully Open, Vision-Centric Exploration of Multimodal LLMs. In: *Advances in Neural Information Processing Systems 38: Annual Conference on Neural Information Processing Systems 2024*; 2024. .
- 37 Zhang S, Xu Y, Usuyama N, Xu H, Bagga J, Tinn R, et al. Biomedclip: a multimodal biomedical foundation model pretrained from fifteen million scientific image-text pairs. *arXiv preprint arXiv:230300915*. 2023.
- 38 Radford A, Kim JW, Hallacy C, Ramesh A, Goh G, Agarwal S, et al. Learning transferable visual models from natural language supervision. In: *International conference on machine learning*. PmLR; 2021. p. 8748-63.
- 39 Guo Z, Xu R, Yao Y, Cui J, Ni Z, Ge C, et al. LLaVA-UHD: An LMM Perceiving Any Aspect Ratio and High-Resolution Images. In: *Computer Vision - ECCV 2024 - 18th European Conference*. vol. 15141; 2024. p. 390-406.
- 40 Grattafiori A, Dubey A, Jauhri A, Pandey A, Kadian A, Al-Dahle A, et al. The Llama 3 herd of models. *arXiv preprint arXiv:240721783*. 2024.
- 41 Wang WY, Wang Z, Suzuki H, Kobayashi Y. Seeing is Understanding: unlocking Causal Attention into Modality-Mutual Attention for Multimodal LLMs. *arXiv preprint arXiv:250302597*. 2025.

- 42 Zhou C, Yu L, Babu A, Tirumala K, Yasunaga M, Shams L, et al. Transfusion: Predict the next token and diffuse images with one multi-modal model. *arXiv preprint arXiv:240811039*. 2024.
- 625 43 Xie J, Mao W, Bai Z, Zhang DJ, Wang W, Lin KQ, et al. Show-o: One single transformer to unify multimodal understanding and generation. *arXiv preprint arXiv:240812528*. 2024.
- 44 Lowekamp BC, Chen DT, Ibáñez L, Blezek D. The design of SimpleITK. *Frontiers in neuroinformatics*. 2013;7:45.
- 45 Brinjikji W, Diehn F, Jarvik J, Carr C, Kallmes D, Murad M, et al. MRI findings of disc degeneration are more prevalent in adults with low back pain than in asymptomatic controls: a systematic review and meta-analysis. *American Journal of Neuroradiology*. 2015;36(12):2394-9.
- 630 46 Lambin P, Leijenaar RT, Deist TM, Peerlings J, De Jong EE, Van Timmeren J, et al. Radiomics: the bridge between medical imaging and personalized medicine. *Nature reviews Clinical oncology*. 2017;14(12):749-62.
- 47 Hu EJ, Shen Y, Wallis P, Allen-Zhu Z, Li Y, Wang S, et al. LoRA: low-Rank Adaptation of Large Language Models; 2021. Available from: <https://arxiv.org/abs/2106.09685>.
- 635 48 Despotović I, Goossens B, Philips W. MRI segmentation of the human brain: challenges, methods, and applications. *Computational and mathematical methods in medicine*. 2015;2015(1):450341.
- 49 Li C, Wong C, Zhang S, et al. LLaVA-Med: training a large language-and-vision assistant for biomedicine in one day. *arXiv. arXiv preprint arXiv:230600890*. 2023.
- 50 Kim Y, Jeong H, Chen S, Li SS, Lu M, Alhamoud K, et al. Medical hallucinations in foundation models and their impact on healthcare. *arXiv preprint arXiv:250305777*. 2025.
- 640 51 Amini A, Vieira T, Ash E, Cotterell R. Variational best-of-n alignment. *arXiv preprint arXiv:240706057*. 2024.
- 52 Wan G, Wu Y, Chen J, Li S. Reasoning aware self-consistency: Leveraging reasoning paths for efficient LLM sampling. In: *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*; 2025. p. 3613-35.
- 645 53 Loshchilov I, Hutter F. Decoupled weight decay regularization. *arXiv preprint arXiv:171105101*. 2017.
- 54 Wu C, Zhang X, Zhang Y, Wang Y, Xie W. Towards generalist foundation model for radiology by leveraging web-scale 2D&3D medical data. *arXiv preprint arXiv:230802463*. 2023.
- 55 Blankemeier L, Cohen JP, Kumar A, Van Veen D, Gardezi SJS, Paschali M, et al. Merlin: a vision language foundation model for 3D computed tomography. *Research Square*. 2024:rs-3.
- 650 56 Papineni K, Roukos S, Ward T, Zhu WJ. BLEU: a method for automatic evaluation of machine translation. In: *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*; 2002. p. 311-8.
- 57 Zhao W, Wu C, Zhang X, Zhang Y, Wang Y, Xie W. RaTEScore: a metric for radiology report generation. *arXiv preprint arXiv:240616845*. 2024.
- 655 58 Delbrouck JB, Chambon P, Chen Z, Varma M, Johnston A, Blankemeier L, et al. RadGraph-XL: a large-scale expert-annotated dataset for entity and relation extraction from radiology reports. In: *Findings of the Association for Computational Linguistics ACL 2024*; 2024. p. 12902-15.
- 59 Lundberg SM, Lee SI. A unified approach to interpreting model predictions. *Advances in neural information processing systems*. 2017;30.
- 660 60 Qiu Z, Xie Z, Lin H, Li Y, Ye Q, Wang M, et al. Learning co-plane attention across MRI sequences for diagnosing twelve types of knee abnormalities. *Nature Communications*. 2024;15(1):7637.
- 61 Suter Y, Knecht U, Valenzuela W, et al. The LUMIERE dataset: longitudinal Glioblastoma MRI with expert RANO evaluation. *Sci Data*. 2022;9:768. Available from: <https://doi.org/10.1038/s41597-022-01881-7>.

- 62 Filimonova E, Pashkov A, Borisov N, Kalinovskiy A, Rzaev J. Utilizing the amide proton transfer technique to
665 characterize diffuse gliomas based on the WHO 2021 classification of CNS tumors. *The Neuroradiology Journal*.
2024;37(4):490-9.
- 63 Zolotova SV, Golanov AV, Pronin IN, Dalechina AV, Nikolaeva AA, Belyashova AS, et al.. Burdenko's Glioblas-
toma Progression Dataset (Burdenko-GBM-Progression) (Version 1); 2023. [https://doi.org/10.7937/](https://doi.org/10.7937/E1QP-D183)
E1QP-D183. The Cancer Imaging Archive.
- 670 64 Boxerman JL, Zhang Z, Safriel Y, Larvie M, Snyder BS, Jain R, et al. Early post-bevacizumab progression on
contrast-enhanced MRI as a prognostic marker for overall survival in recurrent glioblastoma: results from the
ACRIN 6677/RTOG 0625 Central Reader Study. *Neuro-oncology*. 2013;15(7):945-54.
- 65 Gerstner ER, Zhang Z, Fink JR, Muzi M, Hanna L, Greco E, et al. ACRIN 6684: assessment of tumor hypoxia in
newly diagnosed glioblastoma using 18F-FMISO PET and MRI. *Clinical Cancer Research*. 2016;22(20):5079-86.
- 675 66 Ocaña-Tienda B, Pérez-Beteta J, Villanueva-García JD, Romero-Rosales JA, Molina-García D, Suter Y, et al. A
comprehensive dataset of annotated brain metastasis MR images with clinical and radiomic data. *Scientific data*.
2023;10(1):208.
- 67 Snoek L, van der Miesen M, van der Leij A, Beemsterboer T, Eigenhuis A, Scholte S. "AOMIC-ID1000". Open-
Neuro; 2021.
- 680 68 Snoek L, van der Miesen M, van der Leij A, Beemsterboer T, Eigenhuis A, Scholte S. "AOMIC-PIOP1". Open-
Neuro; 2020.
- 69 Silva S, Lorenzi M, Ayed SS. IXI Sample Dataset; 2022. Mendeley Data, V3.
- 70 Rogers CS, Jones MS, McConkey S, Peelle JE. "Listening task". OpenNeuro; 2022.
- 71 Sunavsky A, Poppenk J. "Neuroimaging predictors of creativity in healthy adults". OpenNeuro; 2020.
- 685 72 JE P, B S, MS J, S M, J M, S H, et al.. "Visual and audiovisual speech perception associated with increased
functional connectivity between sensory and motor regions". OpenNeuro; 2022.
- 73 Bilder R, Poldrack R, Cannon T, London E, Freimer N, Congdon E, et al.. "UCLA Consortium for Neuropsychi-
atric Phenomics LA5c Study". OpenNeuro; 2018.
- 74 Nugent AC, Thomas AG, Mahoney M, Gibbons A, Smith J, Charles A, et al.. "The NIMH Healthy Research
690 Volunteer Dataset". OpenNeuro; 2025.
- 75 Park D, Hennessee J, Smith ET, Chan M, Katen C, Wig G, et al.. "The Dallas Lifespan Brain Study". OpenNeuro;
2024.
- 76 Strike LT, Hansell NK, Miller JL, Chuang KH, Thompson PM, de Zubicaray GI, et al.. "Queensland Twin Adoles-
cent Brain (QTAB)". OpenNeuro; 2022.
- 695 77 Zepeda Castellanos J Cpp. Case study, Radiopaedia.org (Accessed on 18 Jun 2025);. [https://doi.org/10](https://doi.org/10.53347/rID-208152)
.53347/rID-208152.

S12. Supplementary Tables and Figures

Supplementary Table 1 | Characteristics of the retrospective primary dataset (Xiangya Hospital) and external test datasets from 11 independent medical centers.

	No. of subjects	Sex		Age in years (range)
		Male	Female	
Total	10,062	4,860 (48%)	5,202 (52%)	53±16 (1–86)
Train dataset	4,936	2,571 (52%)	2,365 (48%)	43±19 (1–86)
Test dataset	5,126	2,289 (45%)	2,837 (55%)	48±17 (1–86)
Xiangya Hospital	8,733	4,198 (48%)	4,535 (52%)	45±19 (1–86)
Changde First People's Hospital	361	193 (53%)	168 (47%)	55±13 (6–86)
Shanghai Tongji Hospital	104	54 (52%)	50 (48%)	54±16 (5–79)
University of South China Second Hospital	110	57 (52%)	53 (48%)	52±15 (6–80)
Shenzhen Second People's Hospital	52	30 (58%)	22 (42%)	49±15 (8–75)
The Third Xiangya Hospital	103	49 (48%)	54 (52%)	52±17 (7–76)
Jiangxi Provincial People's Hospital	232	107 (46%)	125 (54%)	53±15 (13–82)
Chongqing Traditional Chinese medicine Hospital	42	17 (40%)	25 (60%)	58±12 (26–78)
The First Affiliated Hospital of Nanchang University	97	40 (41%)	57 (59%)	46±17 (20–67)
Hunan Provincial Children's Hospital	103	56 (54%)	47 (46%)	28±17 (1–60)
The First Affiliated Hospital of Lanzhou University	81	31 (38%)	50 (62%)	52±13 (8–76)
The Second Affiliated Hospital of Nanchang University	44	28 (64%)	16 (36%)	55±11 (13–74)

In this table, we reported the age of patients using mean ± std.

Supplementary Table 2 | **Summary of publicly available brain MRI datasets.** The table summarizes a range of datasets, detailing the number of subjects, 2D MRI slices, data modalities (2D/3D), and pathological labels (tumor or healthy). For each dataset, original source information and the specific tasks organized in this study are also provided. Here, “3D Healthy” refers to datasets containing healthy individuals with 3D MRI volumes, while “3D Tumor” refers to datasets containing brain tumor patients with 3D MRI volumes. The PubMed Central dataset includes both normal and brain tumor subjects. For ReMIND dataset, we utilized 108 brain tumor cases, excluding 6 brain disease cases. Please note that, due to policy restrictions (e.g., Radiopaedia), certain portions of the training dataset cannot be shared.

Data Type	Data Resource	No. of subjects	No. of images	Original Information	Organized Tasks
2D web crawling	PubMed Central ²⁷	23,849	116,296	Description	2D MRI slice-text description Tumor classification
2D web crawling	Ctibus ²⁵	150	1,124	Description, Diagnosis	2D MRI slice-text description Tumor classification
2D web crawling	ImageCLEFmedical ^{23,29}	484	4,631	Description	2D MRI slice-text description Tumor classification
2D Kaggle	Br35H ²¹	-	2,137	Diagnosis Segment	2D MRI slice-text description Tumor classification
2D Kaggle	Figshare Brain ²² Tumor Dataset	233	3,064	Diagnosis Segment	2D MRI-report Tumor classification
2D Kaggle	Brain Tumor ²⁴ Classification	-	3,264	Diagnosis	Tumor classification
2D Kaggle	Brain Tumor MRI ²³ images 44 classes	-	4,479	Diagnosis	Tumor classification
3D Tumor	LGG-1p19qDeletion ¹⁷	159	32,829	Diagnosis Segment	3D MRI-report Tumor classification
3D Tumor	BraTS23 ^{1,2,3,4,5,6}	3,263	2,156,589	Diagnosis Segment	3D MRI-report Tumor classification
3D Tumor	ReMIND ^{18*}	108	60,963	Diagnosis metadata	3D MRI-report Tumor classification
3D Tumor	RHUI-GBM ⁷	40	122,835	Diagnosis	Tumor classification
3D Tumor	UPENN-GBM ²⁰	630	622,112	Diagnosis Segment	3D MRI-report Tumor classification
3D Tumor	GLIS-RT ¹⁹	230	83,196	Diagnosis	Tumor classification
3D Tumor	QIN GBM Treatment Response ⁶⁰	54	66,988	Diagnosis	Tumor classification
3D Tumor	LUMIERE ⁶¹	91	460,155	Diagnosis	Tumor classification
3D Tumor	OpenNeuro (Diffuse Gliomas) ⁶²	42	22,412	Diagnosis	Tumor classification
3D Tumor	Brain-Tumor- Progression ¹³	20	46,573	Diagnosis Segment	3D MRI-report Tumor classification
3D Tumor	Radiopaedia ²⁶	2606	376, 374	Diagnosis Report	3D MRI-report Tumor classification
3D Tumor	Burdenko ⁶³	180	460,524	Diagnosis	Tumor classification
3D Tumor	RIDER ¹⁰	19	115,100	Diagnosis	Tumor classification
3D Tumor	ACRIN-DSC- MR-Brain ⁶⁴	123	1,385,512	Diagnosis	Tumor classification
3D Tumor	ACRIN-FMISO- Brain ⁶⁴	50	508,500	Diagnosis	Tumor classification
3D Tumor	Meningioma- SEG-CLASS ⁶⁵	96	83,556	Diagnosis metadata	3D MRI-report Tumor classification

Supplementary Table 2 | **Continuation of Table 2. Summary of publicly available brain MRI datasets.** The table summarizes a range of datasets, detailing the number of subjects, 2D MRI slices, data modalities (2D/3D), and pathological labels (tumor or healthy). For each dataset, original source information and the specific tasks organized in this study are also provided. Here, “3D Healthy” refers to datasets containing healthy individuals with 3D MRI volumes, while “3D Tumor” refers to datasets containing brain tumor patients with 3D MRI volumes. The PubMed Central dataset includes both normal and brain tumor subjects. For ReMIND dataset, we utilized 108 brain tumor cases, excluding 6 brain disease cases.

Data Type	Data Resource	No. of subjects	No. of images	Original Information	Organized Tasks
3D Tumor	Brain metastase MRI dataset ⁶⁶	75	85,400	Diagnosis Segment	3D MRI-report Tumor classification
3D Tumor	Brain-TR-GammaKnife ¹²	47	8,300	Diagnosis	Tumor classification
3D Healthy	AOMIC-ID1000 ⁶⁷	928	1, 101, 652	Healthy	Tumor or healthy classification
3D Healthy	AOMIC-PIOP1 ⁶⁸	216	251, 380	Diagnosis	Tumor or healthy classification
3D Healthy	IXI ⁶⁹	582	398, 774	Diagnosis	Tumor or healthy classification
3D Healthy	OpenNeuro (Listening Task) ⁷⁰	78	136, 694	Diagnosis	Tumor or healthy classification
3D Healthy	OpenNeuro (Healthy Adults) ⁷¹	66	122, 175	Diagnosis	Tumor or healthy classification
3D Healthy	OpenNeuro (Visual Audiovisual Speech) ⁷²	60	105, 160	Diagnosis	Tumor or healthy classification
3D Healthy	OpenNeuro (Dynamic Passive Threat) ⁷²	82	126, 299	Diagnosis	Tumor or healthy classification
3D Healthy	OpenNeuro (UCLA Consortium) ⁷³	272	285, 910	Diagnosis	Tumor or healthy classification
3D Healthy	OpenNeuro (NIMH Healthy) ⁷⁴	157	461, 281	Diagnosis	Tumor or healthy classification
3D Healthy	DLBS dataset ⁷⁵	315	99, 344	Diagnosis	Tumor or healthy classification
3D Healthy	QTAB dataset ⁷⁶	422	1, 251, 467	Diagnosis	Tumor or healthy classification

Supplementary Table 3 | Characteristics of prospective dataset, including surgical group and non-operative group for each brain tumor type.

	Surgical Group				Non-operative group				Entire dataset
	No. of subjects	Sex		Age in years Mean $\pm$ Std (range)	No. of subjects	Sex		Age in years Mean $\pm$ Std (range)	
		Male	Female			Male	Female		
Total	639	290	359	48 $\pm$ 17 (2-83)	74	33	41	41 $\pm$ 21 (2-66)	713
MET	25	16	10	60 $\pm$ 8 (38-72)	7	4	3	45 $\pm$ 11 (34-55)	32
GCT	15	11	6	16 $\pm$ 9 (7-39)	3	2	1	8 $\pm$ 1 (7-9)	18
GGN	119	69	50	44 $\pm$ 19 (5-83)	18	8	10	30 $\pm$ 24 (2-60)	137
MEN	185	44	141	53 $\pm$ 11 (14-78)	22	6	16	50 $\pm$ 11 (29-66)	207
TSR	146	81	68	49 $\pm$ 16 (4-77)	17	9	8	54 $\pm$ 8 (37-63)	163
MNM	40	22	21	38 $\pm$ 21 (2-72)	2	1	1	15 $\pm$ 1 (14-16)	42
CPN	87	36	52	52 $\pm$ 13 (22-76)	5	3	2	54 $\pm$ 15 (42-62)	92
CPT	4	1	3	43 $\pm$ 15 (19-60)	-	-	-	-	4
HEM	10	5	5	64 $\pm$ 7 (54-74)	-	-	-	-	10
EMB	7	5	2	17 $\pm$ 12 (5-35)	-	-	-	-	7
PIN	1	0	1	43	-	-	-	-	1

In this table, we use abbreviations for tumor categories: MET (brain metastases), GCT (germ cell tumors), GGN (gliomas, glioneuronal tumors, and neuronal tumors), MEN (meningioma), TSR (tumors of the sellar region), MNM (mesenchymal, non-meningothelial tumors), CPN (cranial and paraspinal nerve tumors), CPT (choroid plexus tumors), HEM (hematolymphoid tumors), EMB (embryonal tumors), PIN (pineal region tumors), and MEL (melanocytic tumors).

Supplementary Table 4 | Abbreviation of 12 major brain tumors in WHO CNS5.

Acronym	Original Tumor Type Name
GGN	Cliomas, glioneuronal tumors, and neuronal tumors
MET	Metastasis
GCT	Germ cell tumor
MEN	Meningioma
TSR	Tumor of the sellar region
MNM	Mesenchymal, non-meningothelial tumor
CPN	Cranial and paraspinal nerve tumor
CPT	Choroid plexus tumor
HEM	Hematolymphoid tumor
EMB	Embryonal tumor
PIN	Pineal tumor
MEL	Melanocytic tumor

Supplementary Table 5 | **Overview of additional MRI sequences and patient information utilized in radiological diagnosis in the primary test dataset and external test dataset.** Extra MRI sequences refer to imaging techniques beyond standard T1, T1c, T2, and FLAIR. These extra sequences include Diffusion-Weighted Imaging (DWI), Diffusion Tensor Imaging (DTI), Perfusion-Weighted Imaging (PWI), Magnetic Resonance Angiography (MRA), Magnetic Resonance Spectroscopy (MRS), Susceptibility-Weighted Imaging (SWI), Magnetic Resonance Venography (MRV), Blood Oxygen Level Dependent Imaging (BOLD), Computed Tomography (CT), and Apparent Diffusion Coefficient (ADC). Additional patient information includes access to medical history and prior radiology imaging and associated findings, offering further clinical context for diagnostic decision-making.

Dataset	Total Number	Extra Modality Percentage	Extra Prior Medical History of Pathological diagnosis	Overall Extra Information Percentage
Primary test dataset	3,797	39.5%	19.3%	58.8%
External dataset	1,309	56.1%	12.7%	68.8%

Supplementary Table 6 | Summary of MRI scanner manufacturers, models, and scanning parameters used across all medical centers.

Scanning Protocol	MRI Scanner Manufacturer			
	GE	Siemens	Philips	UIH
Model	Discovery MR750w, Signa HDxt, Signa Excite	Skyra, Prisma, Avanto, Syngo	Ingenia, Ingenuit	uMR 660
Field strength	1.5 T/ 3.0 T	1.5 T/ 3.0 T	3.0 T	1.5 T
Flip angle	1°, 8°, 12°, 15°, 18°, 19°, 20°, 25°, 30°, 40°, 60°, 70°, 90°, 111°, 112°, 125°, 142°, 155°, 160°	7°, 8°, 9°, 10°, 15°, 20°, 24°, 25°, 27°, 30°, 40°, 50°, 60°, 70°, 80°, 90°, 120°, 122°, 140°, 145°, 150°, 160°, 180°	8°, 10°, 15°, 17°, 18°, 20°, 30°, 40°, 45°, 70°, 75°, 80°, 90°, 100°	10°, 12°, 15°, 20°, 25°, 54°, 60°, 70°, 72°, 90°, 120°, 130°, 140°, 150°
Pixel Spacing Range (mm^2)	0.17×0.17 - 3.5×3.5	0.23×0.23 - 4.7×4.7	0.21×0.21 - 1.8×1.8	0.26×0.26 - 1.4×1.4
Slice Thickness Range (mm)	0.6 - 278	0.47 - 60	0.75 - 398	0.5-10

Supplementary Table 7 | **The designed special tokens for data differentiation and task-driven prompt design to distinguish between different tasks.** A 2D MRI slice is enclosed with '<image>' and '</image>' tags, while 3D MRI volumes use '<vid>' and '</vid>' tags. The and <vi> special tokens represent the 2D image features and 3D volume features, respectively. Additionally, a special token '<s>' is inserted between frames to aid in learning spatial dependencies within the 3D volume.

Tasks	Visual Tokens	Task-specific Instructions Example	Sample Output
2D MRI slice level tumor classification	<image></image>	Is a tumor present in this MRI slice? If so, what type?	<tumor_i>
2D MRI slice-text description	<image></image>	Please make a description for this image.	A lesion in left lobe, showing T1 hyperintense
3D MRI tumor classification	<vid1><v1><v1>...</vid1> <vid2><v2><v2>...</vid2> ...	For a 6-year-old male patient, is a tumor present in these MRIs? If so, what type?	<tumor_i>
3D MRI-report generation	<vid1><v1><v1>...<vid1><s> <vid2><v2><v2>...</vid2><s> ...	For a 6-year-old male patient, please generate a radiology report for him.	In the ..., hyperintense... midline shift to ...
3D MRI report generation tumor classification	<vid1><v1><v1>...</vid1><s> <vid2><v2><v2>...</vid2><s> ...	For a 6-year-old male patient, please generate a radiology report and make a diagnosis.	In the ..., hyperintense...; <tumor_i>
Uncertainty quantification	<vid1><v1><v1>...</vid1><s> <vid2><v2><v2>...</vid2><s> ...	For a 6-year-old male patient, please generate a radiology report, make a diagnosis, and generate reliable score for your diagnosis.	In the ..., hyperintense... midline shift to ... <tumor_i>, confidence: 80%

Supplementary Table 8 | **Diagnostic performance comparison between full MRI sequences (T1, T1c, T2, FLAIR) input and excluding one of the four standard sequences, evaluated on 3,571 patients from the primary and external test datasets.** Only patients with complete MRI data across all four sequences were included to eliminate potential confounding due to missing sequences. The remaining 1,555 patients in the original primary and external test datasets with missing sequences were excluded from the analysis.

	F_1 score				
	Full Sequence	Excluded T1	Excluded T1c	Excluded T2	Excluded FLAIR
MET	0.58	0.52	0.45	0.54	0.53
GCT	0.64	0.48	0.47	0.58	0.53
GGN	0.83	0.80	0.72	0.82	0.82
MEN	0.92	0.89	0.82	0.89	0.89
TSR	0.75	0.65	0.52	0.63	0.63
MNM	0.56	0.43	0.44	0.45	0.48
CPN	0.81	0.76	0.64	0.79	0.78
CPT	0.44	0.27	0.27	0.43	0.27
HEM	0.58	0.43	0.42	0.42	0.45
EMB	0.67	0.44	0.4	0.42	0.44
PIN	0.43	0.36	0.22	0.43	0.29
MEL	0.17	0.17	0.17	0.17	0.17
Frequency-weighted F_1	0.82	0.78	0.71	0.79	0.79

In this table, we use a series of abbreviations to replace original categories: MET for brain metastases, GCT for germ cell tumors, GGN for gliomas, glioneuronal tumors, and neuronal tumors, MEN for meningioma, TSR for tumors of the sellar region, MNM for mesenchymal, non-meningothelial tumors, CPN for cranial and paraspinal nerve tumors, CPT for choroid plexus tumors, HEM for hematolymphoid tumors, EMB for embryonal tumors, PIN for pineal region tumors and MEL for melanocytic tumors.

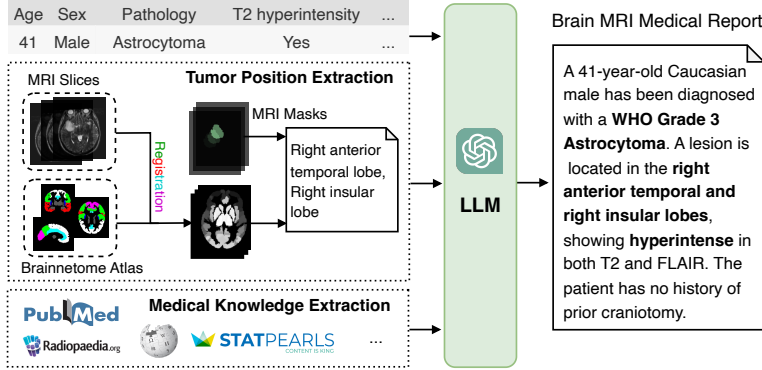
Supplementary Table 9 | Comparison of F_1 scores between our BrainVLM model, neuroradiologists, and three state-of-the-art AI models (RadFM,⁵⁴ Merlin,⁵⁵ and ChatGPT-4o³³). We evaluated 3,797 patients from the primary test dataset and 1,329 patients from the external test dataset, respectively.

	F_1 score (Primary Testing)					F_1 score (External Testing)				
	Ours	Doctor	RadFM ⁵⁴	Merlin ⁵⁵	GPT-4o ³³	Ours	Doctor	RadFM ³⁸	Merlin ⁵⁵	GPT-4o ³³
MET	0.53	0.56	0.45	0.25	0.04	0.61	0.52	0.44	0.40	0.10
GCT	0.57	0.51	0.33	0.15	0.08	0.73	0.52	0.30	0.10	0.11
GGN	0.83	0.82	0.68	0.59	0.47	0.8	0.70	0.45	0.56	0.38
MEN	0.91	0.91	0.64	0.73	0.61	0.82	0.83	0.53	0.63	0.41
TSR	0.88	0.88	0.75	0.49	0.56	0.87	0.84	0.39	0.55	0.48
MNM	0.56	0.43	0.30	0.12	0.03	0.46	0.55	0.27	0.0	0.0
CPN	0.78	0.74	0.40	0.27	0.17	0.8	0.74	0.28	0.30	0.11
CPT	0.54	0.51	0.19	0.36	0.08	0.29	0.18	0.0	0.0	0.0
HEM	0.64	0.45	0.31	0.15	0.0	0.41	0.39	0.33	0.28	0.0
EMB	0.68	0.58	0.47	0.29	0.13	0.48	0.42	0.26	0.25	0.0
PIN	0.41	0.11	0.27	0.17	0.0	0.36	0.17	0.0	0.0	0.0
MEL	0.22	0.12	0.0	0.0	0.0	-	-	-	-	-
Weighted F_1	0.83	0.81	0.62	0.54	0.37	0.76	0.71	0.43	0.52	0.30

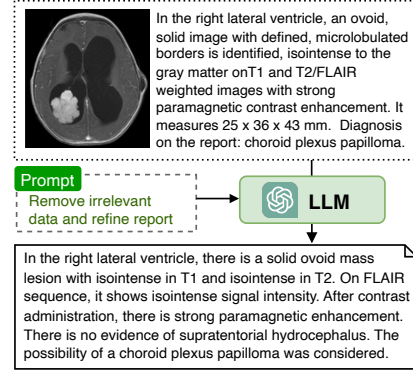
In this table, we use a series of abbreviations to replace original categories: MET for brain metastases, GCT for germ cell tumors, GGN for gliomas, glioneuronal tumors, and neuronal tumors, MEN for meningioma, TSR for tumors of the sellar region, MNM for mesenchymal, non-meningothelial tumors, CPN for cranial and paraspinal nerve tumors, CPT for choroid plexus tumors, HEM for hematolymphoid tumors, EMB for embryonal tumors, PIN for pineal region tumors and MEL for melanocytic tumors.

a. Pipeline for Processing Structured Repositories and Pre-cured Datasets

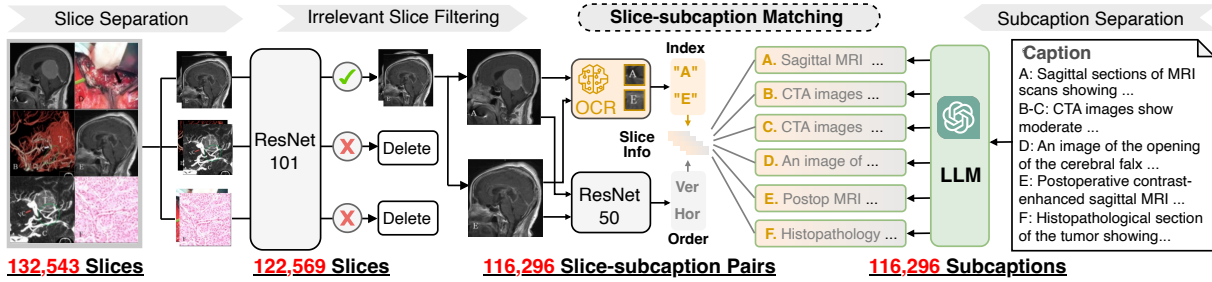
a1. Datasets with Tumor Masks



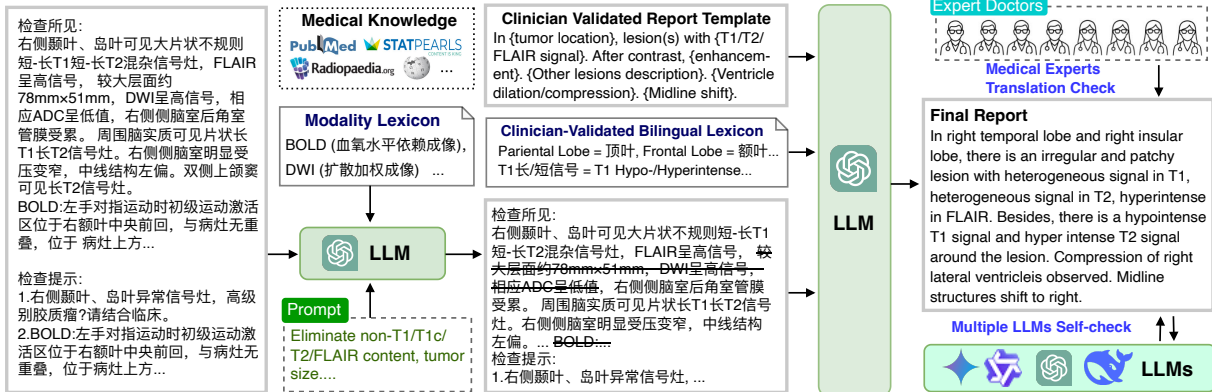
a2. Datasets with Raw Radiology Report



b. Pipeline for Processing Unstructured Web-based Resources

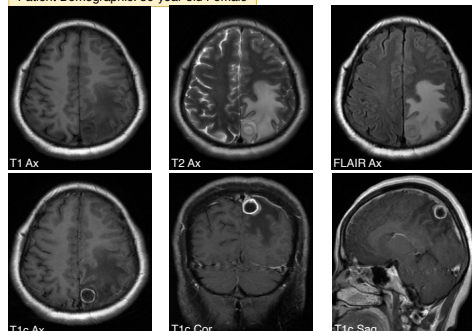


c. Pipeline for Processing Private Medical Institutions



Supplementary Figure 1 | **Preprocessing pipeline for BrainTumor46K.** **a.** The preprocessing pipeline for structured repositories and pre-cured datasets. **a1:** For pre-cured datasets containing patient metadata and tumor masks, multi-sequence MRIs and their corresponding segmentation masks were co-registered to a standard neuroanatomical atlas to localize tumors. All available metadata—including imaging modality, gender, diagnostic labels, and extracted tumor descriptions—were utilized to generate comprehensive medical reports using ChatGPT-4o, prompted with RAG techniques. **a2:** For datasets comprising multi-parametric MRI scans and accompanying radiology reports(Radiopaedia), irrelevant information was removed from the original reports using prompt-based filtering with ChatGPT-4o.⁷⁷ **b.** Preprocessing workflow for unstructured web-based resources, such as PubMed. Using PubMed Central as an example, we illustrate the preprocessing strategy for extracting MRI slice-subcaption pairs. First, brain tumor MRI figures were retrieved through keyword search, and irrelevant images were filtered using a ResNet-101 classifier. Second, compound images were decomposed into individual slices with a Faster R-CNN detector. Third, subcaptions were associated with image slices using Google OCR and ChatGPT-4o. The numbers of MRI slices and resulting slice-subcaption pairs are also summarized below. **c.** The preprocessing pipeline for institutional medical data. Chinese radiology reports were trimmed to retain modality-specific information, reformatted to the standard template, translated into English, and quality-checked by clinicians and multiple large-language models.

Patient Demographic: 50-year-old Female



Significant Omission

Model Prediction with Confidence Score and Radiology Report

Findings:

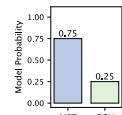
In left parietal lobe, there is a nodular lesion with **hypointense** (Missing of description of isointense) in T1, **hyperintense** in T2, **hyperintense** in FLAIR. After contrast administration, there is a **significant enhancement**. Compression of the left lateral ventricle is observed. Midline structures shift to the right.

Impression:

A nodular lesion was observed in **left parietal lobe**, consider the possibility of **brain metastase tumor** (Confidence 75%).

Gold Standard Pathology Labels (Left parietal lobe)

Brain Metastase tumor



Significant Error

Radiologist Report

Finding: In left parietal lobe, there is a mass lesion with hypointense-isointense in T1, hyperintense signal in T2 and FLAIR. After contrast administration, there is a significant ring-like enhancement. Compression of the left lateral ventricle is observed. Midline structures shift to the right.
Impression: A mass lesion in the parietal lobe, consider **brain metastase tumor**.

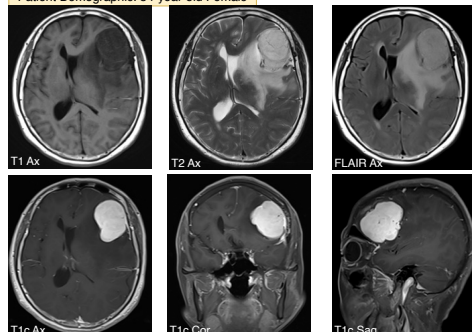
GPT-4o Generated Report

Finding: In the **left frontal lobe**, there is a well-circumscribed mass lesion demonstrating **hypointensity** on T1-weighted images and **hyperintensity** on T2-weighted/FLAIR sequences. After contrast administration, there is uniform **enhancement**. Midline shift to the left.
Impression: A mass lesion in the **left frontal lobe**, consider **meningioma**.

Radfm Generated Report

Finding: In **left parietal lobe**, there is a mass lesion with **hypointense-isointense** in T1, **heterogenous** signal in T2, **hyperintense** in FLAIR. After contrast administration, there is **significant enhancement**. Midline structures **shift to the right**. (Missing of compression description)
Impression: A mass lesion was observed in **left parietal lobe**, considering **meningioma**.

Patient Demographic: 54-year-old Female



Model Prediction with Confidence Score and Radiology Report

Findings:

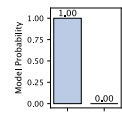
In **left frontal lobe**, there is a mass lesion with **hypointense** in T1, **hyperintense** in T2, **hyperintense** in FLAIR. After contrast administration, there is a (homogeneous) **significant enhancement**. Compression of the left lateral ventricle is observed. Midline structures shift to the right.

Impression:

A mass lesion was observed in the **left frontal lobe**, considering the possibility of **meningioma**. (Confidence 100%).

Gold Standard Pathology Labels (Left frontal lobe)

Meningioma



Radiologist Report

Finding: In left frontal lobe, there is a mass lesion with hypointense in T1, hyperintense in T2, hyperintense in FLAIR. After contrast administration, there is a homogeneous enhancement. Compression of the lateral ventricles is observed. Midline structures shift to the right.
Impression: A mass lesion in the left frontal lobe, consider **mesenchymal, non-meningothelial tumour**.

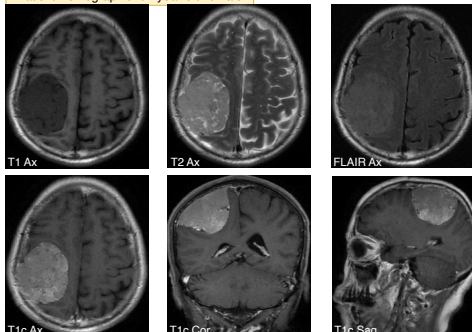
GPT-4o Generated Report

Finding: In the **right frontal region**, there is a well-circumscribed extra-axial mass lesion demonstrating **hypointensity** in T1 and **hyperintensity** in T2, with **homogeneous contrast enhancement**. Surrounding vasogenic edema is evident as **hyperintensity** on FLAIR images. Midline shift is **not noted**. Ventricular obstruction is **not noted**.
Impression: A mass lesion in **right frontal region**, consider **meningioma**.

Radfm Generated Report

Finding: In **Right frontal lobe**, there is a mass lesion with **hypointense** in T1, **hyperintense** in T2, **hyperintense** in FLAIR. After contrast administration, there is a (homogeneous) **marked enhancement**. No **significant dilatation** of the ventricular system. **No midline structure shift**.
Impression: A mass lesion in **right frontal lobe**, consider **brain metastase tumor**.

Patient Demographic: 67-year-old female



Model Prediction with Confidence Score and Radiology Report

Finding

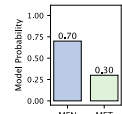
In **right parietal lobe**, there is a mass lesion with **hypointense** in T1, **hyperintense** in T2, **hyperintense** in FLAIR. After contrast administration, there is a **significant enhancement**. Compression of the right lateral ventricle is observed. Midline structures shift to the left. This patient was diagnosed with Meningioma.

Impression:

A mass lesion was observed in the **right parietal lobe**. Consider the possibility of **meningioma** (Confidence 70%) and brain metastase tumor (Confidence 30%).

Gold Standard Pathology Label (Right parietal lobe)

Meningioma



Radiologist Report

Finding: In right parietal lobe, there is a mass lesion with hypointense in T1, hyperintense in T2, hyperintense in FLAIR. After contrast administration, there is a marked enhancement. Compression of the right lateral ventricle is observed. Midline structures shift to the left.
Impression: A mass lesion in the right parietal lobe, consider **meningioma**.

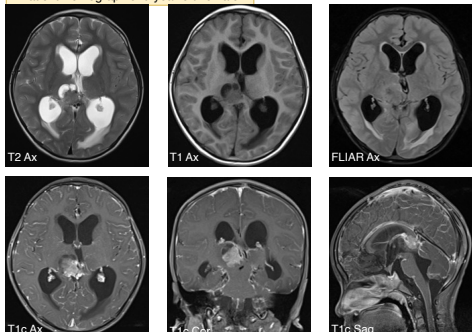
GPT-4o Generated Report

Finding: In **left temporal lobe**, there is a mass lesion with **hypointense** signal in T1, **hyperintense** signal in T2, **hyperintense** signal in FLAIR. After contrast administration, there is **heterogeneous** significant enhancement. Compression of the **right lateral ventricle** is observed. **No midline shifts**.
Impression: A mass lesion found in **left temporal lobe**, consider **glioma**.

Radfm Generated Report

Finding: In **right frontal lobe**, there is a mass lesion with **hypointense** in T1, **hyperintense** in T2, **hyperintense** in FLAIR. After contrast administration, there is a **marked enhancement**. No **significant dilatation** of the ventricular system. **No midline structure shift**.
Impression: A mass lesion was observed in **right frontal lobe**, consider **brain metastase tumor**.

Patient Demographic: 6-year-old female



Model Prediction with Confidence Score and Radiology Report

Findings:

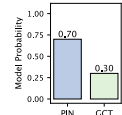
In **pineal region**, there is a mass lesion with **hypointense** in T1, **hyperintense** in T2, **hyperintense** in FLAIR. After contrast administration, there is a **significant enhancement** of the solid component. Dilation of bilateral ventricles is observed. (Compression of the third ventricle is observed). No midline structure shift.

Impression

A mass lesion was observed in **pineal region**, considering the possibility of **pineal region tumor** (Confidence 70%) and germ cell tumor (Confidence 30%).

Gold Standard Pathology Label (Pineal region)

Pineal region tumor



Radiologist Report

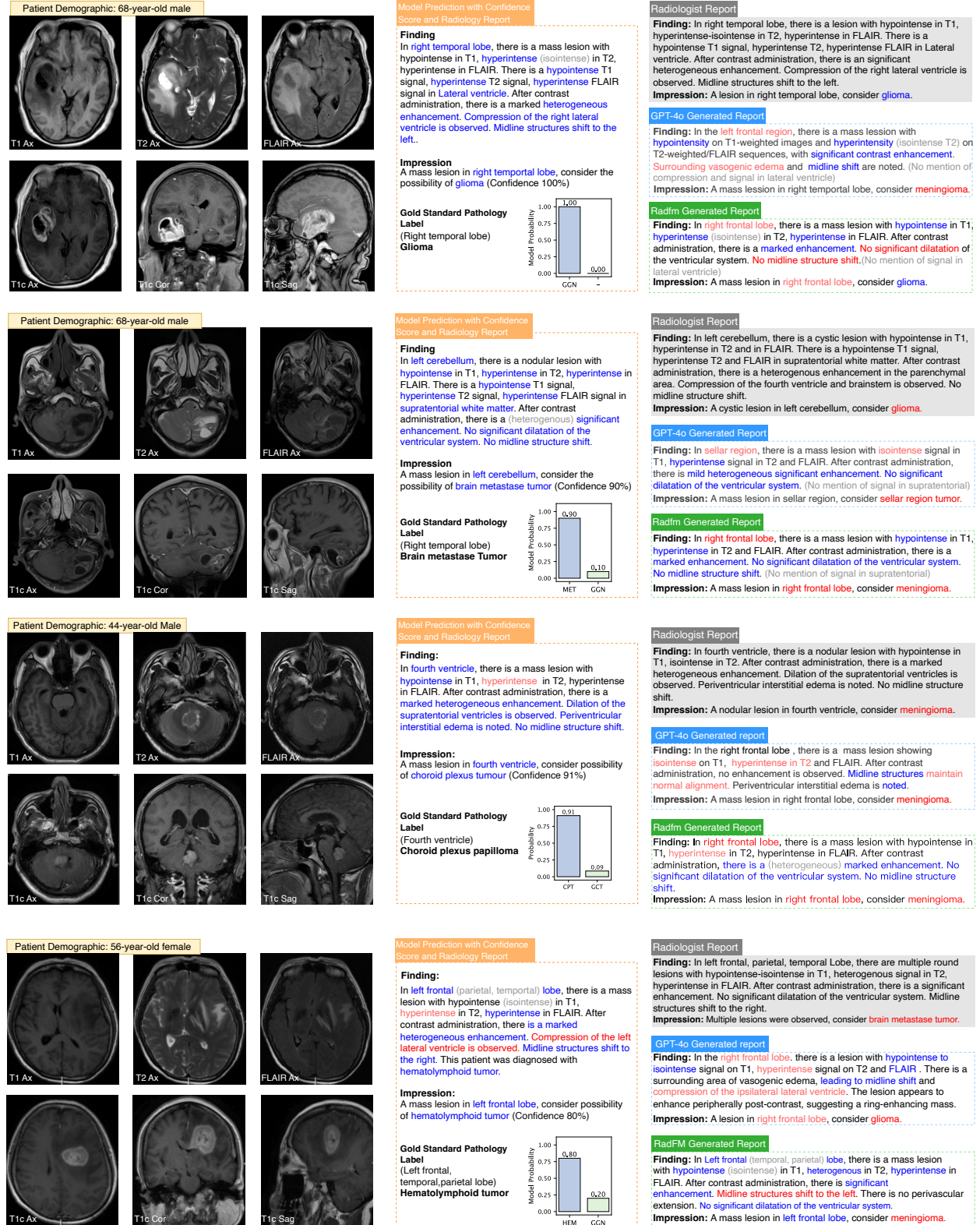
Finding: In right pineal region, there is a lesion with hypointense in T1, hyperintense in T2, hyperintense in FLAIR. After contrast administration, there is a significant enhancement of the solid component. Dilation of bilateral ventricles is observed. Compression of the third ventricle is observed.
Impression: A lesion located in right pineal region, consider **germ cell tumor**.

GPT-4o Generated Report

Finding: In the **occipital horns of the lateral ventricles**, there is a mass lesion with **hypointense** signal in T1, **hyperintense** signal in T2 and FLAIR. After contrast administration, there is **significant enhancement**. No midline shifts.
Impression: A mass lesion in the **lateral ventricles**, consider **glioma**.

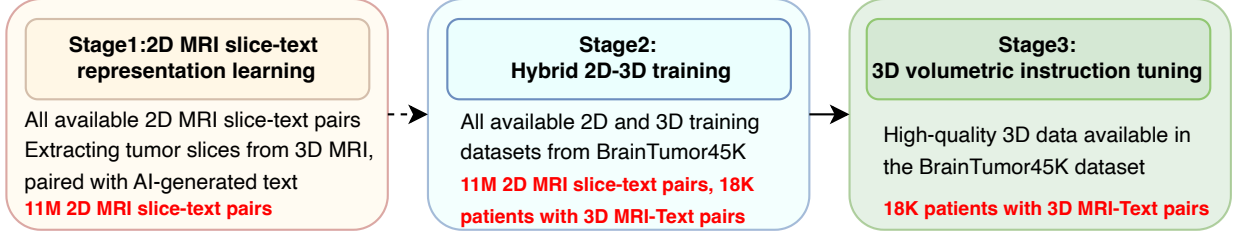
Radfm Generated Report

Finding: In **right frontal lobe**, there is a mass lesion with **hypointense** in T1, **hyperintense** in T2, **hyperintense** in FLAIR. After contrast administration, there is a **marked enhancement**. No **significant dilatation** of the ventricular system. **No midline structure shift**.
Impression: A mass lesion in the **right frontal lobe**, consider **germ cell tumor**.

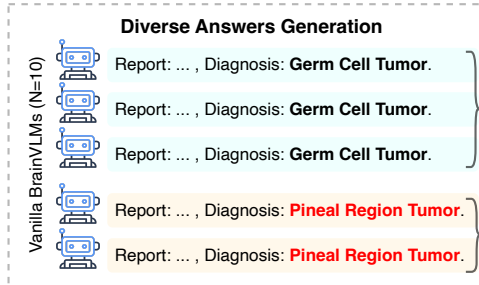


Supplementary Figure 2 | Examples of reports and diagnoses generated by our BrainVLM, compared with those from doctors (expert-curated radiology report) and other state-of-the-art AI models (RadFM⁵⁴ and ChatGPT-4o³³). For each diagnosis generated by BrainVLM, a corresponding reliability score is also provided. We use a series of abbreviations to replace original categories: MET for brain metastases, GCT for germ cell tumors, GGN for gliomas, glioneuronal tumors, and neuronal tumors, MEN for meningioma, TSR for tumors of the sellar region, MNM for mesenchymal, non-meningothelial tumors, CPN for cranial and paraspinal nerve tumors, CPT for choroid plexus tumors, HEM for hematomphoid tumors, EMB for embryonal tumors, PIN for pineal region tumors and MEL for melanocytic tumors.

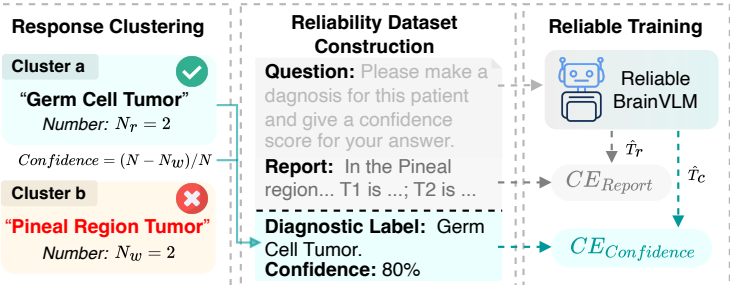
a. Three-stage progressive training strategy of BrainVLM



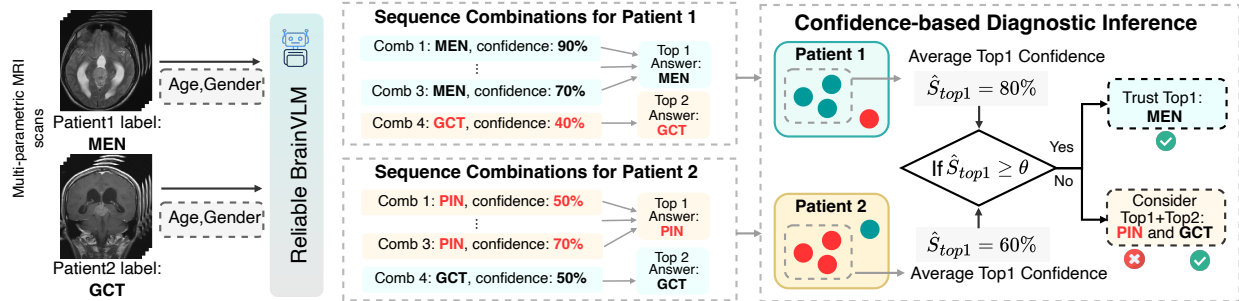
b. Consensus-driven Response Clustering



c. Diagnostic Reliability Training

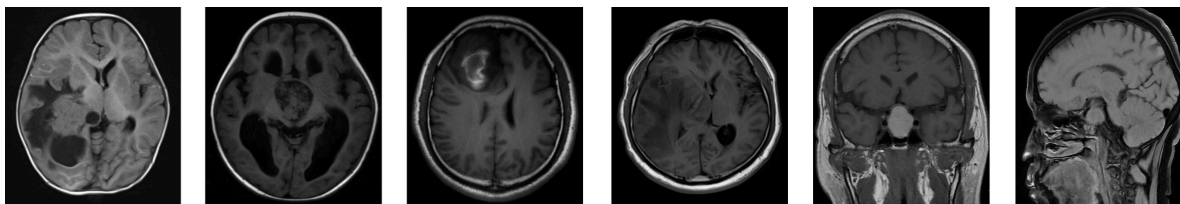


d. Reliability-driven Inference of Top2 Diagnostic Candidates

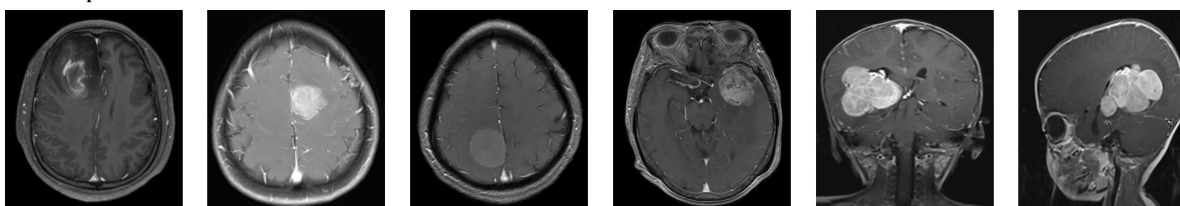


Supplementary Figure 3 | **Three-stage progressive training strategy and uncertainty quantification implementation of BrainVLM, including the confidence-triggered Top 2 supplementary diagnosis inference pipeline.** **a. Three-stage progressive training strategy of BrainVLM.** BrainVLM was developed using a three-stage progressive training approach: (1) 2D MRI slice-text representation learning using all curated 2D MRI slice-text pairs; (2) volumetric context integration via hybrid 2D-3D training with all available 2D and 3D training datasets; and (3) comprehensive 3D volumetric instruction tuning leveraging the high-quality 3D data in BrainTumor46K. **b,c, Uncertainty quantification implementation pipeline.** The uncertainty quantification of our BrainVLM consists of two steps: consensus-driven reliability dataset construction and confidence-aware reliable model finetuning. **b. Consensus-driven reliability dataset construction.** A reliability dataset is constructed by evaluating N vanilla BrainVLM models on the training dataset and clustering their diagnostic answers to derive confidence scores. **c. Confidence-aware reliable BrainVLM finetuning.** The consensus-derived reliability knowledge in the reliability dataset is distilled into reliable BrainVLM through confidence-aware reliable fine-tuning. **d. Confidence-triggered Top 2 supplementary diagnosis inference.** During deployment, each prediction made by the reliable BrainVLM is accompanied by a confidence score. If the confidence score of the Top 1 prediction falls below a predefined threshold (set at 75% in this study), the Top 2 prediction is also reported as a supplementary diagnosis to enhance clinical robustness. In this figure, we use abbreviations for tumor categories: MET (brain metastases), GCT (germ cell tumors), GGN (gliomas, glioneuronal tumors, and neuronal tumors), MEN (meningioma), TSR (tumors of the sellar region), MNM (mesenchymal, non-meningothelial tumors), CPN (cranial and paraspinal nerve tumors), CPT (choroid plexus tumors), HEM (hematolymphoid tumors), EMB (embryonal tumors), PIN (pineal region tumors), and MEL (melanocytic tumors).

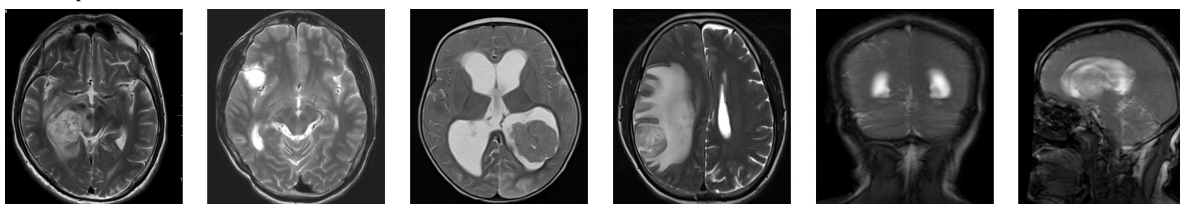
a. T1 Comparison



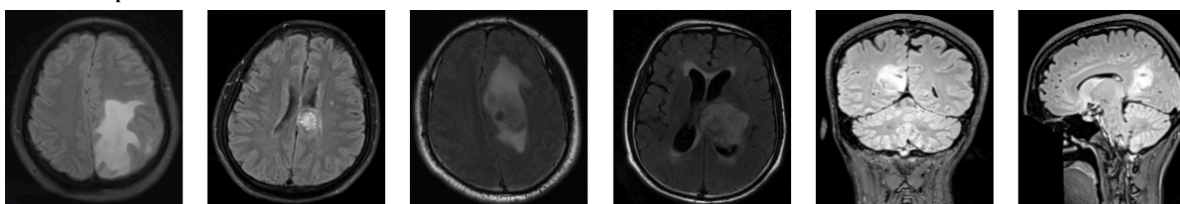
b. T1c Comparison



c. T2 Comparison

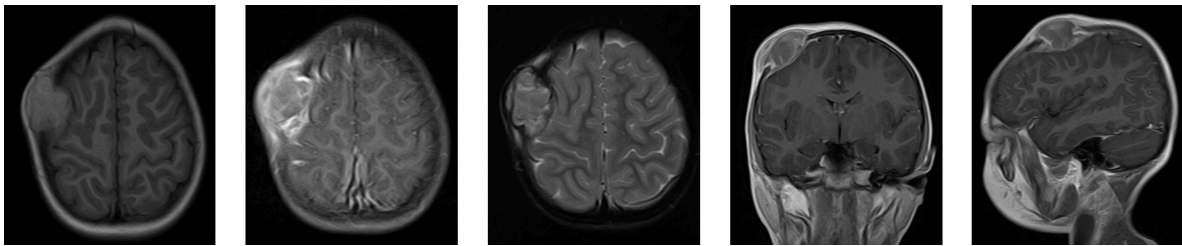


d. FLAIR Comparison

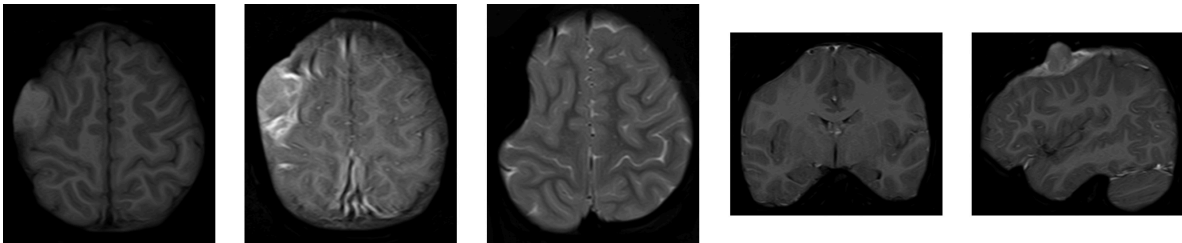


Supplementary Figure 4 | Visualization of MRI images acquired using different imaging parameters across MRI sequences. Shown are T1-weighted (T1), contrast-enhanced T1-weighted (T1c), T2-weighted (T2), and T2 FLAIR (FLAIR) sequences. For each sequence, four axial images obtained using different imaging parameters and MRI devices are presented. Additionally, one representative sagittal and one coronal view acquired under a specific parameter setting are included for each sequence.

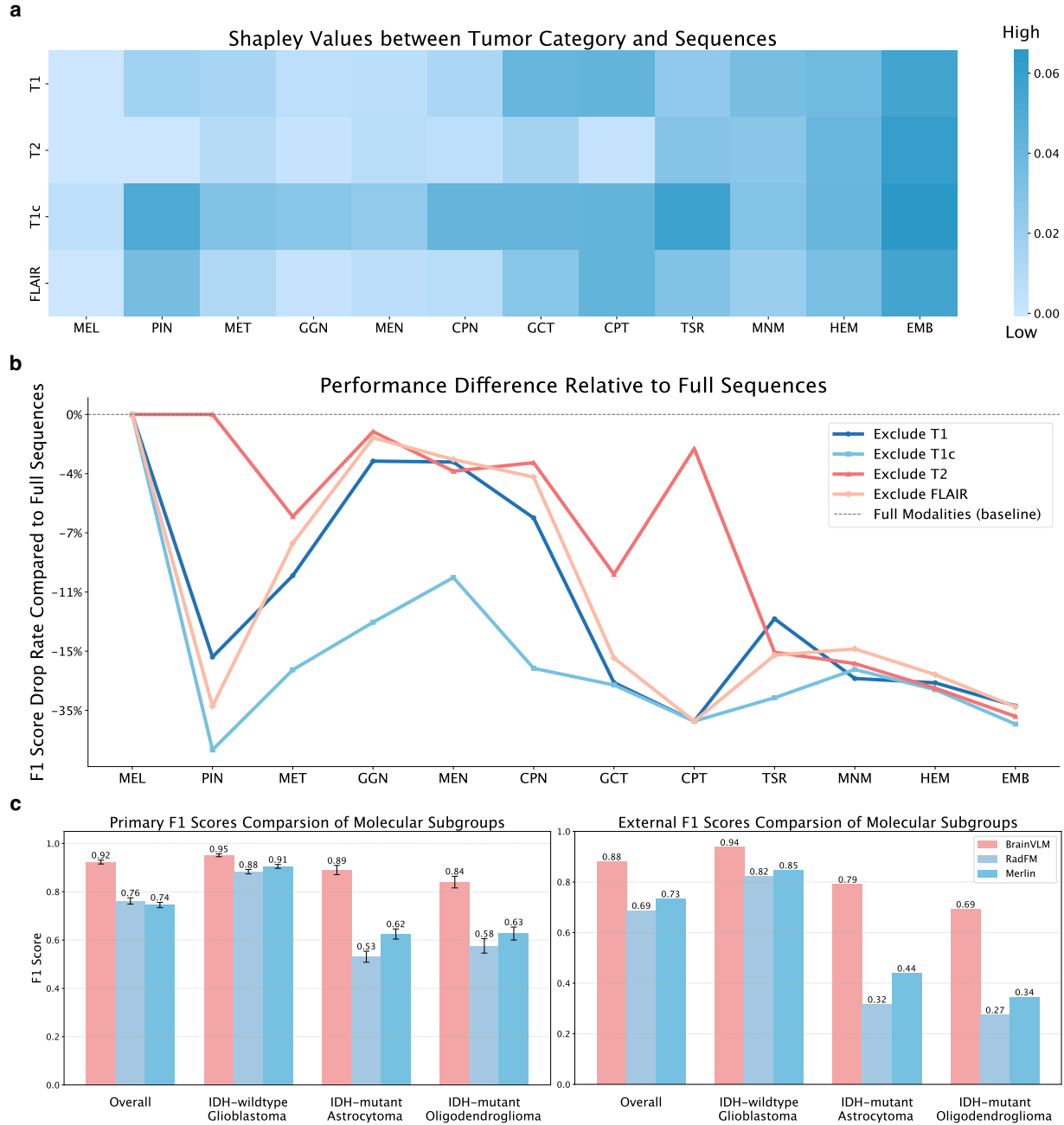
a. Original MRI image



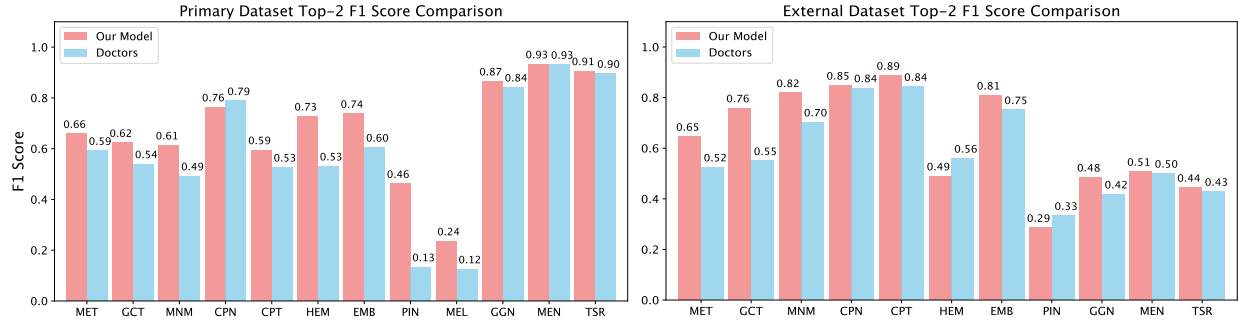
b. MRI image after skull-stripping



Supplementary Figure 5 | **Comparison of unprocessed and skull-stripped MRI images.** Representative examples of MRI scans before and after skull-stripping. While conventional skull-stripping may aid certain tumor analyses, it is not universally appropriate. Crucially, lesions located in regions like the pia mater or invading the cranial vault can be inadvertently removed during skull stripping, leading to the loss of critical anatomical and pathological information – a risk our approach deliberately avoids.



Supplementary Figure 6 | **Impact of individual MRI sequences on BrainVLM diagnostic accuracy for brain tumor classification.** Diagnostic performance was evaluated in 3,571 patients by comparing the full-sequence model (T1, T1c, T2, FLAIR) to models in which each sequence was individually omitted. Only patients with complete MRI data across all four sequences were included to eliminate potential confounding due to missing sequences. The remaining 1,555 patients in the original primary and external test datasets with missing sequences were excluded from the analysis. For each tumor type, the performance drop rate (measured in F_1 score) resulting from the exclusion of a specific modality is shown (**b**, central panel). Both the direct performance drop rate and Shapley value analysis (**a**, top panel) highlight distinct sequence dependencies among tumor classes: T1c removal resulted in the greatest performance decline across all tumor types, underscoring its essential role. EMB classification was uniformly sensitive to the absence of any sequence (35% mean drop), while GGN and MEN predictions remained stable when T2 or FLAIR or T1 was missing. In contrast, CPT predictions were more dependent on T1, T1c and FLAIR, but showed resilience to the exclusion of T2. **c**, Molecular subgroups F1 scores comparison between our BrainVLM with Merlin and RadFM. In this figure, we use a series of abbreviations to replace original categories: MET for brain metastases, GCT for germ cell tumors, GGN for gliomas, glioneuronal tumors, and neuronal tumors, MEN for meningioma, TSR for tumors of the sellar region, MNM for mesenchymal, non-meningothelial tumors, CPN for cranial and paraspinal nerve tumors, CPT for choroid plexus tumors, HEM for hematolymphoid tumors, EMB for embryonal tumors, PIN for pineal region tumors and MEL for melanocytic tumors.



Supplementary Figure 7 | F_1 score comparison across 12 tumor types, evaluating the Top 2 predictions of our model and neuroradiologists on both primary and external datasets. In this figure, we use abbreviations for tumor categories: MET (brain metastases), GCT (germ cell tumors), GGN (gliomas, glioneuronal tumors, and neuronal tumors), MEN (meningioma), TSR (tumors of the sellar region), MNM (mesenchymal, non-meningothelial tumors), CPN (cranial and paraspinal nerve tumors), CPT (choroid plexus tumors), HEM (hematolymphoid tumors), EMB (embryonal tumors), PIN (pineal region tumors), and MEL (melanocytic tumors).